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Examine the potential of Artificial Intelligence in enhancing project management agility: Concerning Norwegian Companies.

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This thesis is worth 30 study points.

Abstract

Systems capable of learning and accomplishing tasks in the same way as the human brain works are known as Artificial intelligence (AI). It can change and improve various industries, including supply chain, management, medicine, and automobiles, by changing business models, improving performance, and enhancing individual skills. Project management is a method used to accomplish a successful outcome at the end of the project. During the past ten years, the growth of AI in project management has offered excellent reasons for expected accomplishments, primarily employed in the assessment of project progress, evaluation, decision-making, and prediction. This research investigates how AI technologies can make the project management process more agile. A mixed-method approach was used to arrive at a more accurate conclusion at the end of the research. It has taken nearly three months to gather the required data through a survey and interviews with the project managers who work in Norwegian companies. This study was conducted in Norway. The hypothesis and data that were gathered were analyzed using Microsoft Excel and the Statistical Package for Social Science (SPSS).

The four variables discussed are AI integration, Organizational learning culture, Change management practices, and Project management agility. A frequency analysis was conducted to analyze the demographic profiles of the project managers who participated in the survey. Reliability and descriptive analysis were performed to analyze the data gathered from the research. Regression and thematic analysis have been undertaken to prove the hypothesis and address research questions. The findings of this study show that AI integration, Organizational learning culture, and change management practices are associated with greater project management agility.

Keywords: Project management, Artificial intelligence, Management, Information systems, AI integration, Project management agility, Organizational learning culture and Change Management practices

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Vishmi Senarath

1 Introduction

1.1 Background Study

Agile development is a process that replaces traditional methods, which heavily depend on many documents (Hussein, 2009). Agile Methods take the place of the traditional software development methodology that was heavily based on documents (Juricek, 2014). It is a new method of managing and organizing software projects (Dybå et al., 2014). It has experienced an increase in popularity over the years because of its success in fulfilling various changing demands through stages and creating value (Ansar, 2024). The approaches that project managers have used in the past to implement their projects with traditional methods have been successful in their delivery with expected results. However, the traditional approaches fail to fulfill the ongoing requirement changes, and hence, the desired objective cannot be attained (Wysocki, 2011). This can be addressed by the use of advanced technologies such as AI in project management procedures (Tominc et al., 2023).

AI includes technology that allows one to accomplish jobs that previously required human intellect (Jarrahi, 2018). Integrating AI has the potential to revolutionize project management by improving flexibility, efficiency, and response to changing conditions (Daraojimba et al., 2024). Agile approaches encourage cooperation, openness, and continual development (Daraojimba et al., 2024). Furthermore, AI dramatically improves project management agility by automating repetitive processes. Divide project tasks such as distributing resources, planning, and predicting possible hazards can be done effectively with the use of AI-driven tools (Shamim & Islam, 2024). Other than that, Shamim and Islam (2024) have investigated that there is the possibility of evaluating prior project data and forecast hazards and detecting particular patterns that enable project managers to deal with the problems regarding time duration and development process. Also, the chatbots powered by AI and digital assistants improve interaction and transparency by providing immediate information to stakeholders all over the project lifespan.

Furthermore, AI-powered analytics enable project managers to have a greater awareness of the project's success and make real-time data-driven choices. Project teams may use machine learning algorithms to detect undetected patterns, recognize trends, simplify project operations, and increase productivity (Alfaifi & Aksoy, 2023). AI algorithms can do this by assigning the right tasks to suitable teammates based on their duties, areas of expertise, and availability. Another benefit is the fact that resources are assigned according to the nature of the task; every member can ease their process and be able to complete the given tasks at a higher level (Alfaifi & Aksoy, 2023). The latest technology makes it very simple to communicate with project stakeholders through chatbots and virtual assistants (Vijaya et al., 2024). AI technologies dedicated to that task will facilitate communication between individuals by making it transparent and reliable. Therefore, they can send their messages to one another and console each other from anywhere and anytime. This makes the stakeholders more involved, and thus, they can work more effectively to reach their common goal by increasing the cooperation of the team (Vijaya et al., 2024).

Several empirical studies support the potential of AI to enhance agility in project management. For instance, research done by Soravito (2023) specifies that predictive analysis will help to detect the possible dangers in the project, and it will lead to enhancements in the result of the project (Soravito, 2023). Bainy (2024) discovered that it is possible to minimize any delays regarding the project while increasing communication among team members by using AI-powered chatbots (Bainey, 2024).

Different determinants influence the enhancement of agility in project management. Organizational learning culture can be identified as a condition that helps employees in an organization to learn something in a smooth way (Rebelo & Duarte, 2011). Moreover, resultantly, it enhances agility within the project management context. With reference to Jarrahi et al. (2022), they have discussed the influence of AI on organizational culture. Change management is an activity that comes after any shift or transition has been established in an organization (Naz et al., 2013). Further, they have done a study about the implications of using AI technologies to Change management practices by creating a requirement change management model.

1.2 Problem Statement

There is a constant demand for project managers to balance conflicting priorities and handle several jobs at once. The main issue regarding project management is the inadequate support of the organization's upper management to project managers (Bhavsar et al., 2019). AI is essential in project administration (Aghlamazyan, 2022), urging and inspiring personnel to undertake and monitor projects at all stages (Bhavsar et al., 2019). A broader team's engagement in the project makes it necessary to understand how to incorporate stakeholders to improve results (Bhavsar et al., 2019). Collaboration among stakeholders on projects at work has grown increasingly important for the project's success when a more extensive team becomes involved (Bhavsar et al., 2019). They assign specific tasks to an intelligent helper who never complains about being weary and is capable of analyzing vast volumes of information, identifying trends, drawing conclusions from them, and producing extraordinarily fast forecasts (Aghlamazyan, 2022).

Insufficient project management and outdated technology can lead to project failure or jeopardize the anticipated software project output (Bhavsar et al., 2019). Not only that, using spreadsheets, presentations, and other software can also lead to the failure of a project. Even though these are older methods, some companies still use these traditional ways to manage their projects (Nieto-Rodriguez & Vargas, 2023). This will be good when project success is determined by meeting timelines, but those are insufficient when projects are continuously altering and evolving the business (Nieto-Rodriguez & Vargas, 2023). There are some AAI-based project management applications such as Ayanza, Trello, and Wrike, but there still needs to be more automation, organizing capacity for group communication, and other elements.

Considering the widely acknowledged possible advantages of AI in project management, several critical questions and challenges remain unresolved. These include reviewing the success of AI integration in improving project outcomes, comprehending the impact of human-AI cooperation, dealing with ethical issues about applying AI, deciding organizational preparedness for AI adoption, and guaranteeing the sustained success of powered AI strategies. Tackling these concerns is critical for deepening our knowledge of AI's function in project management and providing advice for organizations on how to leverage AI to promote agility and stimulate creativity successfully.

Looking back on prior research, some studies focus on the utilization of AI in project management, but they need to talk specifically about Norway. Also, previous research has only addressed particular PM issues by concentrating on specific AI applications. However, the papers currently available about choosing technology and assessment do not adequately take into account the peculiarities of AI (Johnk et al., 2020). The total of research conducted between the years 2005 to 2015, the number of studies on AI remained very low in the years 2007, 2008, 2010, and 2012. There was a significant increase in AI-related studies in 2019 and 2020, indicating a higher level of interest in the topic (Collins et al., 2021). Although project managers claim that AI technologies are increasing the efficiency of projects and improving the caliber of work, there are no investigations that address this issue and concentrate on analyzing AI approaches in the various areas of project management (Taboada et al., 2023).

Taboada et al. (2023) findings indicated a notable rise over the past ten years in the number of articles on artificial intelligence-enabled project management. Bley et al. (2022) found out there is a positive relationship between Organizational culture, AI capabilities, and Organizational performance. Timo et al. (2021) studied the effect of Machine learning, which is an AI technology, on organizational learning. A paper by Engel, Ebel, and Giffen (2021) examines the relationship between attributes of AI, project management, and organizational change. One study has discussed the utilization of AI technology in management and further investigated change management practices needed to successfully adopt AI in an organization (Dixit & Mishra, 2024). AI application in project management was examined by Davahli (2020), who evaluated 652 articles from several databases and concluded that there are still a lot of unrealized prospects in the early phase of AI use in project management (Davahli, 2020).

Due to the lack of research on the application of AI technology in project management, Bodea, Mitea, and Stanciu (2020) took part in an international study conducted by IMPA (International Project Management Association) and PwC Romania in 2020, with the primary goal of determining the perceived status of AI adoption in project management. Jahrrahi et al. (2022) proposed a framework that shows AI’s contribution to organizational culture but does not address anything regarding project management. A higher number of examined research concentrated on using AI to improve scheduling and planning, but there is a gap in the research, nevertheless, since the AI models were not utilized in projects in project management (Nagireddy, 2023).

In Norway, AI applications are being considered and deployed in project management to help speed up the process and ensure success. Even so, there is still a vast area of knowledge yet to be investigated on how to incorporate AI technologies into project management practices. Previous research could propose concepts regarding AI adoption, like AI's influence on organizational culture or performance. However, there is just a minimal body of empirical research concentrating on the operationalization of AI tools and methods within the Norwegian project management context. Apart from that, the social and organizational dynamics in Norway, which are so unique and specific in the culture and organization in Norway, make it necessary to study how AI can be used to solve the challenges and opportunities in Norwegian project management. There is a clear need for research work that exhibits how AI could optimize PM performance, as well as assessing factors unique to the Scandinavian environment that influence its adoption and implementation within Norwegian organizations. Therefore, the current research problem statement can be developed as follows.

How can AI be utilized to enhance project management agility in the context of Norwegian organizations?

1.3 Purpose Statement

To address the recognized gap in the literature, this investigation is trying to find the integration of AI for project management in the Norwegian context by getting a sample of project managers who work in Norwegian companies. A mixed-method research design, which combines both quantitative and qualitative designs, has been used to investigate the subject area thoroughly. AI integration of Project management is discussed by identifying factors, including AI integration, Organizational learning culture, and Change management practices.

1.4 Research Objectives

Primary Objective

To investigate the utilization of AI in project management inside organizations based in Norway.

Secondary Objectives

Objective 1: To find out whether there is an impact of AI integration on project management agility in Norwegian companies.

Objective 2: To find out whether there is an impact of organizational learning culture on project management agility in Norwegian companies.

Objective 3: To find out whether there is an impact of change management practices on project management agility in Norwegian companies.

1.5 Research Questions

Primary Research Question

What is the extent of AI utilization in project management within organizations based in Norway?

The main research question is to understand the degree of AI usage and the extent of AI adoption and application in project management processes in organizations that are working in Norway. The study tries to evaluate how much AI technologies are used in project management from the planning phase to the execution, monitoring, and decision-making phases. Through the examination of the extent and scope of AI use in project management in Norwegian companies, this research intends to furnish an understanding of the present AI adoption situation, uncover the possible challenges and opportunities, and evaluate the effect of AI on project management in the Norwegian situation.

Secondary Research Questions

RQ01: Does AI integration impact project management agility in Norwegian companies?

This research question is on the connection between AI and project management in Norwegian companies. The study aims to find out if the use of AI technologies makes the project management processes in the organizations that operate in Norway more agile, i.e., they become more reactive to changes, adaptable, and able to perform overall projects better. Through the analysis of the degree of the AI integration effect on project management agility, this study aims to provide awareness of how AI technologies are tools for the improvement of project management efficiency and effectiveness in the Norwegian situation.

RQ02: Does organizational learning culture impact project management agility in Norwegian companies?

The given question concentrates on investigating the impact of the organizational learning culture on project management agility in Norwegian companies. It concentrates on the aspects of the organizational culture of continuous learning, knowledge sharing, and adaptability that are responsible for the flexibility of the project management processes. Through the analysis of the connection between the learning culture and project management agility, this research aims the discovery the intensity of the creation of a suitable learning environment that fosters the innovation, flexibility, and responsiveness of the project management practices in Norwegian organizations.

RQ03: Does change management practices impact project management agility in Norwegian companies?

The research question investigates the connection between change management practices and project management agility of Norwegian companies. It is to find out how the change management strategies that are successful in the project management processes help in the fastness of the processes of the project management, especially in the provision of the adaptation to the changing project requirements, the expectations of the stakeholders, and the external factors. Through this study, the researchers aim to find out the way of change management practices that are related to project management agility and thus, the fundamental practices and approaches that are promoters of agility and resilience in project management for Norwegian contexts.

2 Theoretical Framework and Literature Review

2.1 Introduction

This chapter contains literature related to the current research context on the variables of AI integration, Organizational learning culture, Change management practices, and Project management agility. Basically, the current research literature review has been done by using the existing literature about theories, principles, and concepts regarding the current research study. Basically, more than 30 articles within the field of project management have been reviewed.

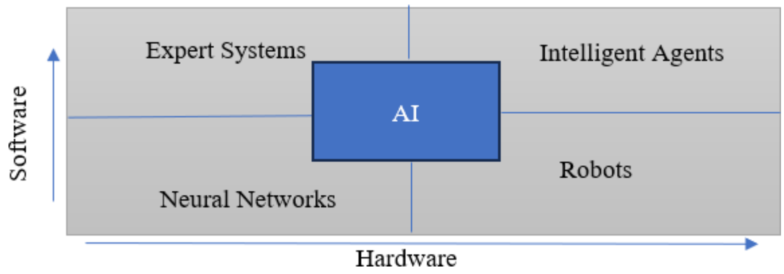
2.2 Artificial Intelligence

2.2.1 AI as an Emerging Technology

There is no doubt that AI is capable of advancing the degree of thinking in computer systems (Nagireddy, 2023; Mahmood et al., 2023) and focuses on programming computers to perform tasks that people are currently more adaptable to (Auth et al., 2019b). AI can be described as the study of concepts that allow new technology to accomplish several tasks that people want to do manually, such as planning, learning, thinking, interaction, and the capacity to navigate and handle some objects (Saini, A., 2023). Researching AI adoption is becoming more vital, and several investigations have attempted to explore the reasons for hurdles, problems, and effects connected with AI applications in companies (Hashfi & Raharjo, 2023). Breakthroughs in computational power, widespread data, and breakthroughs in machine learning have accelerated the industrialization of AI solutions in a variety of industries (Johnk et al., 2020).

According to the research done by Naggireddy (2023), AI can be described as a technology that can complete tasks that humans are unable to complete without cognitive ability. This concept has an enormous possibility to increase the relationship between people and technology. Otero-Mateo et al. (2022) have provided a suitable diagram to illustrate the idea of AI, as shown in the below figure.

Figure 1 Classification and adaption of AI.



The above figure describes AI in 4 areas: Expert systems, which are the systems that reason; Intelligent agents, which are systems that work rationally; Neural networks – systems that can feel like a human being; and Robots, which are the systems that behave like people (Otero-Mateo et al., 2022). Mahmood, Marzooqi, El, and AlAmeemi, 2023, have described AI as computers that are designed to replicate human intellect and learn similarly to how the human brain functions (Mahmood et al., 2023). These computers are capable of accomplishing several tasks intelligently at once without getting tired by adjusting to a variety of scenarios (Mahmood et al., 2023). Systems that use AI can identify traits, group them, form preferences, and provide estimates about the future by examining an extensive data set (Aghlamazyan, 2022). This enables the revolution of the globe by making changes in the production, manufacturing, and delivery procedures with the use of deep learning (Dam et al., 2019).

AI could be classified into two distinct groups: general AI, which means strong AI, and narrow, which means weak AI (Taylor, 2021). Artificial narrow intelligence (ANI), often known as weak AI, can only specialize in a limited range of tasks, and it can perform a hundred thousand calculations per second if built properly (Nagireddy, 2023). Also, this weak AI is the one that powers modern society, and it is utilized in voice recognition innovations such as Alexa, Siri, and Google Home. The second kind of AI is called artificial super intelligence (ASI) and artificial generic intelligence (AGI), which is strong AI (Nagireddy, 2023). With AGI, a machine would possess awareness and intellect comparable to that of humans and would be able to address issues and make plans (Nagireddy, 2023). Also, AGI is capable of mimicking humans across all levels and, theoretically, performing all tasks that humans can undertake.

Machine learning (ML) applies algorithms and acquires information from data without direct programming. Deep learning (DL) is mainly focused on NN, natural language processing (NLP), automation and robotics, machine vision, knowledge-based systems, neural networks (NN), and applied intelligence are different types of AI technologies (Nagireddy, 2023; Saini, 2023).

When considering a company or an organization, they have different sections such as project management, business administration, and research development. AI can be used for all these sections to increase the efficiency of every part. Not only that, but it can a

It has successfully faced every challenge within various types of industries, such as healthcare, entertainment, and finance. Also, it provides chances to revolutionize existing models regarding businesses and the performance of jobs (Collins et al., 2021). Using AI increases the possibility of computers performing tasks that are now best handled by people (Ko & Cheng, 2007) while making dramatic shifts in company legislation, organizational culture, and efficiency (Bodea et al., 2020).

The increment of interest in the usage of AI is shown in the magnitude of worldwide investment, which the International Data Corporation (IDC) estimates will reach over $98 billion in 2023, more than double the $37.5 billion invested in 2019 (Collins et al., 2021). Monitoring the state of a project and modifying it as a requirement are some primary abilities of AI; reducing the cost of a project and quickly finishing the project with higher standards can be achieved through machine learning (Hashfi & Raharjo, 2023). So, managers are inclined to utilize and place confidence in an AI system when they possess a clear understanding of functionalities and advisory capabilities (Fridgeirsson et al., 2021). Improvements in computational power, abundant data accessibility, and advancements in machine learning, among other factors, have hurried the integration of AI applications across various business sectors (Johnk et al., 2020).

After careful study of these things, researchers have been able to come up with various types of uses, including the detection of faces, picture identification, and more (Wang, 2019). Victor (2023) predicted that data collecting, and storage would grow commonplace and that companies with a lot of data, like financial services, medical care, and scheduling, would profit significantly from AI (Victor, 2023). Furthermore, it provides an affordable and functional approach that may be adjusted to the individual demands of enterprises or organizations (Bhavsar et al., 2019).

2.2.2 Benefits of AI

AI provides a fantastic opportunity for massive data analysis and for doing routine work. AI can identify problems that humans might miss and provide creative and new solutions for the problems (Wang, 2019). The majority of companies pay for an expert service without researching new technologies, which will result in a high project cost. AI will demonstrably offer the optimal solution in the first place, by means of the latest technology and thus at a much lower cost of the project.

AI can assist in recognizing errors and alerting users when intervention is required, as well as being capable of planning and scheduling resources based on the goals of the particular project (Mahmood et al., 2023). AI may also be utilized as a collaborative design tool to assist project management, as demonstrated by the Tangible User Interface system (Mahmood et al., 2023).

2.2.3 Challenges of AI

The more complicated function that a company needs to use, the more amount of data will be required, and it may lead to privacy concerns (Wang, 2019). Many companies do not have a sufficient amount of information needed, and the availability of data resources might also be an issue (Wang, 2019). The rate of accuracy of a result is contingent on the level of competency of the technology that is used to generate the result; depending on the technology which is not mature enough will lead to the failure of a project (Q. Wang, 2019). Sometimes, AI systems could provide inaccurate information owing to insufficient system training (Mahmood et al., 2023).

The improvement of Artificial Intelligence may lead to unemployment, and the consequences of this will cause participants of the project to become concerned. It may have an impact on the reputation of the company and will drop profitability (Q. Wang, 2019). Even though using AI is cost-effective, there might be some risks due to limited or not enough quantity of data for analysis, there is restricted access to needed data, or lack of knowledge in AI can be significant problems (Bhavsar et al., 2019). The adoption of AI into current society necessitates a large amount of data storage and excellent handling of the required information to use it as a value-based technology (Mahmood et al., 2023).

Another challenge of AI is that it can highly affect the future job market, and researchers have already studied how it will be affected; Frey and Osborne were two of the initial to publish their investigation by determining that approximately 47% of US employment has a threat of being automated jobs (Fridgeirsson et al., 2021). Approximately 14% of OECD states are in danger of being 70% automated, and 32% of employees are in danger of slight changes due to computerization ranging from 50% to 70% (Fridgeirsson et al., 2021). Another investigation has mentioned that there is a 35% risk for workers in Finland and a 33% risk for workers in Norway due to the digitalization of jobs (Fridgeirsson et al., 2021).

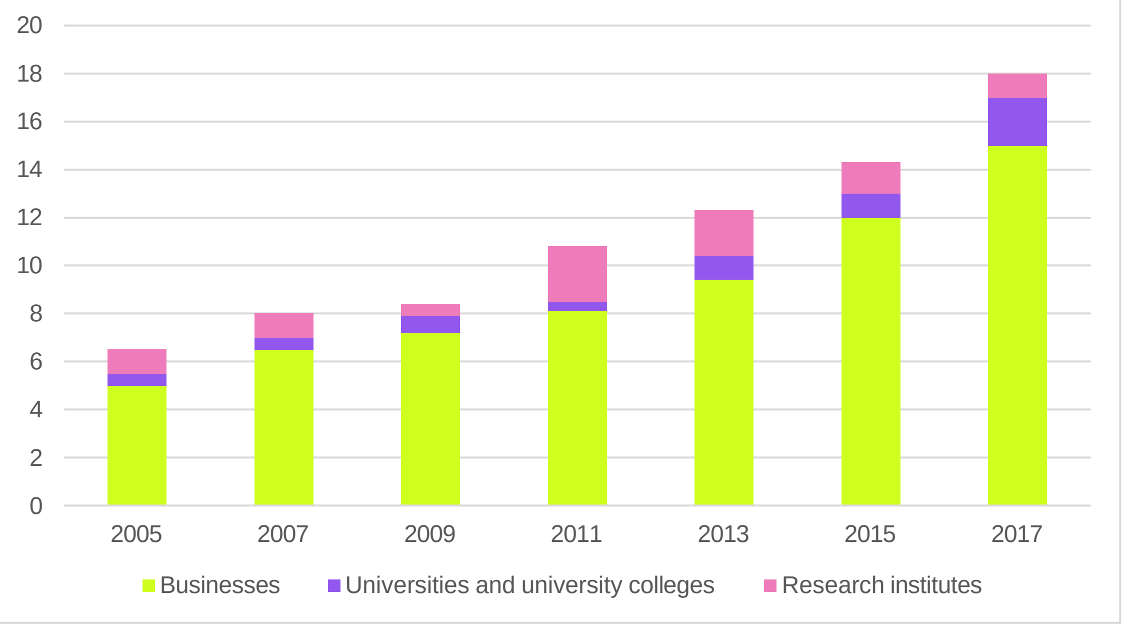
Several studies have investigated bias regarding AI-integrated decision-making. For this process, algorithms are trained using the previous data so they can reinforce pre-existing biases when trained on historically biased data, which may lead to discriminatory practices. For instance, in recruitment, AI-powered tools might inadvertently prefer candidates from certain demographic groups, perpetuating systemic inequalities (Albaroudi et al., 2024). To overcome this type of issue, the best way is to use transparent methods by using different sets of data for evaluations. These actions make sure that AI technologies contribute to fairness. There is a need for more in-depth studies into AI systems that actively combat bias and foster equality in automation (Albaroudi et al., 2024).

2.2.4 AI in Norway

During the last few years, the growth of research in the area of ICT has increased significantly, with a nominal increase from NOK 8 billion to NOK 18 billion between 2007 and 2017. The most significant percentage of R&D investment in ICT is made by industry. Between 2015, and 2most significant largest increase was in the university sector (Modernisation, 2020)

The graph below shows the consumption of IT in Norway (Modernisation, 2020).

Figure 2 - Consumption of IT



The above figure highlights the increasing investment in research and development across a variety of industries. It offers a picture of the shifting landscape of resource allocation among corporations, academic institutions, and research institutes throughout the given time.

2.2.5 Studies Regarding AI

AI focuses on programming computers to perform tasks that people are currently more adaptable to (Auth et al., 2019b). The article by Auth, Jokisch, and Dürk in 2020 discussed how AI has changed over time, how it was used in project management historically, and how interest in it has recently increased. When considering the gaps between existing processes of technology selection and the capabilities of AI, Johnk et al. (2020) have suggested a procedure that takes five steps to create AI use cases. These are very specific to the particular organization. This includes gathering and organizing data on the three context-related aspects, determining current issues and AI remedies, knowledge encouraging abstraction to understand the fundamental nature of problems, linking domain issues with AI answers, and

They are determining the appropriate ramifications for the use cases' eventual implementation (Johnk et al., 2020). The research done by Balcioglu et al. (2022) provides an approach by giving crucial data on employee placement and selection with artificial intelligence. Crawford et al. (2023) examined how AI is used in software engineering and provided an overview; the article mentioned that mistakes committed during the project planning stage are the main reason for the failure of a project, and by using AI, projects can gain a higher success rate.

Nuhn has studied how artificial intelligence systems will impact society and why (Nuhn et al., 2022). Saini talks about the methods and applications of artificial intelligence (Saini, A., 2023). Collins et al. (2021) comprehend the several aspects of AI that have been investigated in Information systems. He also talks about features like speech detection, machine learning, robots, natural language processing, specialist systems, and computer vision (Collins et al., 2021). It is true the fact that AI is growing more and more important across a range of sectors (Gil et al., 2021). As an example, according to another investigation, 56% of respondents stated that AI was a component of their digital transformation strategy. Over 65% percent of those surveyed said their companies had already used AI in one of their regular project management processes or intended to do so over the next three years (Bodea et al., 2020).

2.3 Project Management

The Project Management Body of Knowledge (PMBOK) Guide defines project management as applying expertise, abilities, instruments, and strategies to fulfill the requirements of a project. The application of many techniques, such as predictive, hybrid, and adaptive procedures, paves the way for project teams to accomplish their goals (Clemente & Domingues, 2023). According to Seymour and Hussein's description (2014) in the International Journal of Management and Information Systems, cultivating cooperation is the primary purpose of project management to complete successful projects on time and within budget (Seymour & Hussein, 2014). Overall, project management refers to the use of information, abilities, instruments, and strategies to satisfy specific requirements of a project (Nagireddy, 2023). Project management is vital for making sure that a project is successfully finished with the organization of material needed, supervision of stakeholders, tracking and measuring the progress, and communication effectively inside and outside the organization with project team members and necessary stakeholders (Nagireddy, 2023).

According to the article written by Auth, Jokisch, and Dürk (2019), the ISO 21500 and ANSI/IEE PMI PMBOK international standards have specified ten sub-areas of project management as "integration, stakeholders, scope, human resources, time, cost, quality, risk, procurement, communication" (Auth et al., 2019b). These are incorporated into a five-phase project lifecycle which is about planning, executing, overseeing, regulating, and closure (Auth et al., 2019a). Project management principles include the goal of the project, outcome of the project, cost, And timeline, monitoring, and evaluation of the advancement of the project, as well as hazards (Fotso et al., 2022). A study done by Nagireddy (2023) stated that the four primary elements of project management as time, cost, scope, and quality (Nagireddy, 2023).

Time: Time is an essential component in managing projects, and it refers to the particular project deadline (Butt, 2018). As stated in the book written by Wysocki (2013), the relationship between the length of a project and the estimated expenses for it is vice versa, meaning that cutting the project’s duration would increase the expenditure. Maintaining the timeline and working according to the plan is vital.

Cost: The cost of a project is critical to its success, and completing the project within the estimated cost is a key element (Butt, 2018). Estimation is crucial for the development of the budget and can influence how a project moves forward beyond the planning stage (Butt, 2018).

Scope: The scope of a project outlines its limits and excludes some components (R. Wysocki, 2013). The scope is the most frequently modified part of the project necessitating regular modifications (Butt, 2018).

Quality: There are two types of qualities in a project: the quality of the product and the quality of the process (Wysocki, 2013). Establishing a high-quality project requires pleasing all stakeholders, who may have differing perspectives on success (Butt, 2018).

Project management is not restricted to one business or area; it can be used for initiatives across several sectors, such as the development of a software program, the construction industry, marketing, and healthcare (Nagireddy, 2023). When considering IT projects involve the execution of software, database administration, technical installation, website creation, networking setup, recovery of computer systems, and development of mobile applications (Fotso et al., 2022). The productive implementation of a project is dependent upon the abilities of project managers, assistants, and directors who ensure follow-up and evaluation (Fotso et al., 2022). According to the study done by APM-commissioned and PwC, essential abilities a project manager ought to process are managing and leading others, setting up an expense and control of funds, organizing and tracking, digital skills, and opportunity management (Victor, 2023).

2.3.1 Challenges in Project Management

Improving one’s capacity to conceptualize, design, create, and deploy information systems that match client needs is a critical problem in information systems and technology (Collins et al., 2021). Nevertheless, many projects fail despite the use of project management approaches, and such failures limit the potential of technological components (Collins et al., 2021). Most projects fail because of late delivery, over budget, and lack of planning (Collins et al., 2021).

The article written by Fotso, Pradhan, and Sukdeo (2022) mentioned that firms and organizations approach digital transformation as followers rather than leaders without any commitment to change, which will result in project failure. Unable to reach specified objectives regarding timing and money, moral issues while handling projects, and ineffective communication abilities and collaborations (Nagireddy, 2023).

Another problem within this area is utilizing a vast amount of previously gathered information to optimize the real-time handling of projects (Wang, 2023). Also, project managers are not getting adequate support from the top management of most organizations (Bhavsar et al., 2019).

2.3.2 Agile Project Management

The Agile Project Management Model is mainly an iterative and adaptive method and an essential process in modern project management activities. It highlights mobility, cooperation, and adaptability to shifting needs, which closely correlate with the dynamic nature of today's corporate settings (Rigby et al., 2016). The main objective of this model is to deliver the project outcomes through several sprints. Then, the project team can adapt according to the feedback of the customer (Cockburn & Highsmith, 2001).

Frameworks like Scrum, Kanban, and Extreme Programming (XP) are integral components of the Agile Project Management Model. For instance, Scrum has several sprints during the project, and it encourages team members to organize by themselves. It creates an environment that creates corporations and divides responsibilities among each other (Rupp & Singh, 2020). Kanban is visualizing the project progress using a board which is called a Kanban board. It makes the members of the team divide and manage the work tasks and do it according to a flow (Anderson et al., 2012). XP stresses collaboration, testing, and ongoing integration to achieve excellent outcomes and respond quickly to changing needs (Beck, 1999).

As mentioned earlier, the Agile Project Management Model enables the members of the team to adjust according to changes and challenges within a less amount of time (Serrador & Pinto, 2015). By embracing principles such as customer collaboration, responding to change, and delivering working solutions iteratively, organizations can improve project outcomes and adapt more effectively to uncertain and evolving business environments (Highsmith, 2001). Accordingly, the Agile Project Management Model offers a strong foundation for handling projects in changing and unpredictable environments (Wysocki & K, 2010). Its concepts and practices allow teams to provide value iteratively, work directly with participants, and respond to changing necessities, making it ideal for today's corporate environment.

Agile Project Management is a technique, and being agile necessitates a significant behavioral shift in team members' attitudes and actions inside the company (Loiro et al., 2019). Tesch, Ireland, and Liu (2008) examine the project management efforts made by IS scholars to take into account the covering of essential areas of project management by describing important components of project management such as participation of user's scale of the project and planning (Tesch et al., 2008). To improve project maturity, efficiency, and accomplishment rate, Vayyavur's (2016) study focuses on intelligent project portfolio management systems. A paper by Drabble (1995) described a few concepts that could contribute to the development of the upcoming project management software.

The article by Dam et al. (2019) proposed a structure for integrating and adapting various AI technologies to assist the types of aspects of agile project management by making use of tasks such as sprint planning, risk management, creating user stories, backlog item identification, and refinement will be simpler to complete with AI tools than they will be with human intervention. This study also forms a chatbot that runs on software and serves as a conduit between users and the rest of the AI system.

2.4 Overview of AI in Project Management

2.4.1 AI in Project Management

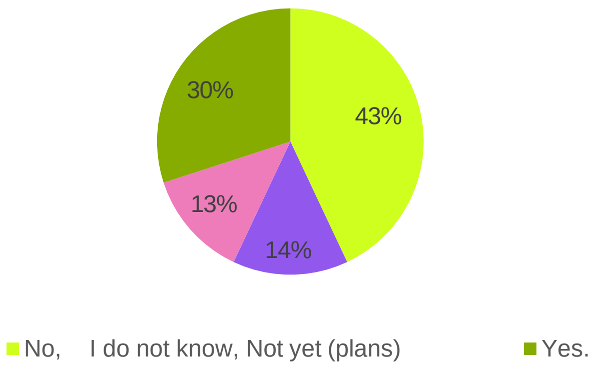
Project management is the application of necessary knowledge and skills to achieve a common goal, and it has been using AI since 1987 (Lahmann, 2018). Shortly, technological developments are going to have a significant influence on project management, and AI will undoubtedly change how project management tasks are performed and handled. AI is the replicating and modeling of a person's mental operations by computers, which enable robots to replicate human skills (Aghlamazyan, 2022). It is a sub-study of computing and technology that highlights the progression of knowledge and machine learning that play out all tasks like regular people. SPM (software project management) is an approach for programming business associations that incorporates project arranging, planning, and asset distribution (Bhavsar et al., 2019). Project managers can get advice from AI-based tools to achieve their project goals, and it leads to receiving the desired outcomes (Taboada et al., 2023).

When considering the previous investigations, project managers spend over fifty percent of their days engaged in bureaucratic tasks, including improvements and monitoring (Victor, 2023). According to the research done by Fotso, Pradhan, and Sukdeo (2022), around 44% of ICT projects fail to satisfy company objectives, 43% of projects are delayed and under budget, and 18% of projects entirely fail owing to a lack of strategic efforts. Zadeh (1996) mentioned that for every $1 billion invested in the United States, $122 million was squandered owing to poor project planning and performance.

To avoid these types of failures, the foremost duty of project managers is to provide an atmosphere where individuals can collaborate to accomplish a shared goal, with the ultimate goal of completing successful projects as scheduled within the specific timeline and the estimated cost (Nagireddy, 2023). Mistakes in collecting requirements for a project, cost estimation, and risk assessment might end up in higher expenses or project failure (Crawford et al., 2023). So, the majority of the issues that occur during a project encountered by project managers can be significantly reduced with the use of technology; project managers cannot resolve everything by themselves (Nagireddy, 2023). As a result, many software developers search some methods whereby AI may be used to decrease the errors associated with project management choices (Crawford et al., 2023). Applying Artificial Intelligence to project management can reduce the percentage of failed projects because AI will substantially alter the method in which they operate their businesses within the coming five years (Taboada et al., 2023).

Predicted by 2030, AI-powered by big data, machine learning, and natural language processing will take care of 80% of project management responsibilities (Nieto-Rodriguez & Vargas, 2023). According to a recent PMI research, 44% of business projects in the sector are using resources and techniques supported by AI, and 70% of commercial organizations plan to begin cooperating with AI in project management (Hazari, 2022). Researchers have differing perspectives on the potential impact of AI on occupation (Fridgeirsson et al., 2021). PWC's 2018 Workforce of the Future Report anticipates a growing demand for project management methodologies and processes that can proceed faster and more flexibly (Nagireddy, 2023). According to the research done by Nagireddy (2023), the market provided $1.5 billion in investment for 200 AI-related startups during the first half of the year 2016, and they mentioned the proportion of AI systems within the companies, as shown in the figure below—only a moderate number of companies used AI in 2023.

Figure 3 - Percentages of AI Systems in the company



AI can positively impact the domain of project management (Mahmood et al., 2023). The article written by Nieto and Vargas (2023), which was published in Harvard Business Review, discusses that managers are currently researching how to apply AI in Project Management (Nieto-Rodriguez & Vargas, 2023). There, the authors illustrated the current way of managing projects and how it will be more beneficial and smoother after applying AI (Nieto-Rodriguez & Vargas, 2023). Project Management is frequently used to develop solutions that function and fulfill the demands of customers (Tesch et al., 2008). Project management is increasingly viewed as a source of competitive advantage in modern industries that rely significantly on it (Soderlund, 2005). The project's features and the selected PM methodology both influence PM operations, and as a result, AI solutions must work with the applicable process model (Ledesma, 2014).

According to Nieto and Vargas (2023), project executives and managers can easily view the latest state of the project and predicted rewards by just touching apps on their cell phones (Nieto-Rodriguez & Vargas, 2023). It says that the app they made uses AI to update everything immediately, and communications are sent to relevant individuals within the team, notifying them of the changes and providing an estimate of the expected outcome.

With the advancement of AI, project managers will increasingly rely on technology to predict upcoming patterns, streamline time administration processes, and address both superior and employee needs more efficiently (Victor, 2023). So, the tasks assigned to project managers can be easily automated, automatically sending messages to required team members, gathering and preserving data, examining past data, and making the required forecast (Nagireddy, 2023). According to Taboada et al. (2023), AI will be included in project management procedures in the years to come and will immediately affect the areas of expertise related to project managers. When AI is connected with project management, a system may manage and administrate projects on its own, eliminating the need for human interaction (Bhavsar et al., 2019). Results of Taboada et al. (2023) study show that the management of cost, scheduling, and risk significantly will be impacted by AI, project size, support requirement specification, and estimate work necessary (Ledesma, 2014). AI may be the ideal answer for tasks requiring a great degree of inventiveness, speed, agility, and originality (Hazari, 2022). Research done by Fridgeirsson et al. (2021) mentioned that one survey investigated and found 1770 managers across 14 countries project managers are more likely to receive guidance from AI systems rather than higher rank managers.

Forecasting can be enabled by machine learning, which additionally guides project managers, and soon, AI will turn mental mappings into a network of terms and use it to extract tasks and their connections (Lahmann, 2018). Software development companies are seeing a rise in the use of AI, and project managers are starting to see its possibilities (Bhavsar et al., 2019). Also, AI assists managers in putting the agile method into practice for operation management (Bhavsar et al., 2019). Stepsize AI, Chatbots, Stratejos, ZiveBox, PolyOne, Trello, Motion, Notion, Monday, ClickUp, and Ayansa are some of the AI tools and software that can be used for projects (Nagireddy, 2023). These tools may assist managers in setting priorities, dividing up urgent problems, and allocating resources wisely so the project management can make sure that resources are not squandered and prevent missing project deadlines (Hazari, 2022). Comparatively speaking to other industries like banking, marketing, and customer services, the majority of global businesses use AI in the IT sector to identify security breaches, complicated technical issues, and internal compliance (Fotso et al., 2022).

One crucial measure of the caliber of the project is to reduce the number of errors that can be detected easily by using AI, and it can enhance project management by offering more perspectives on likely outcomes (Victor, 2023). There will be changes to agile implementation as well. The use of an agile project helper powered by AI can improve the efficiency of projects, as mentioned in the article by Victor (Victor, 2023). Software for speech detection and web searches, such as Google and Siri, among others, use AI concepts to carry out their tasks, and computers may do intelligent activities associated with project management that were previously limited to the project manager's intellect, like problem-solving and establishing decisions (Nagireddy, 2023). Tonchia (2018) claims that AI facilitates the evaluation of the project's outcome, which leads to creative project approaches (Tonchia, 2018). The book by Peter Taylor discusses AI applications in project management, teaching project managers how to automate and enhance data, so there has been some advancement in project management with the emergence of AI (Taylor, 2021).

According to the above literature, the table below can be used to get a clear idea about different types of AI with relevant project management tasks, which is created using the above findings.

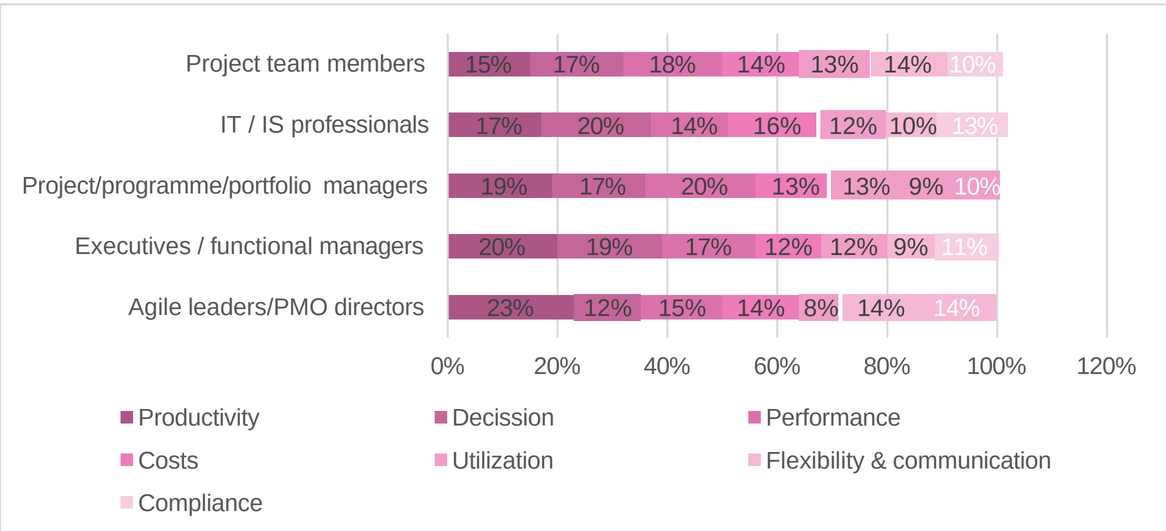
Table 1 - Different types of AI

AI Technology	Project Management Task	Advantages
Predictive analytics	Error detection	• Earlier recognition of possible errors. • Minimizing the risk. • Increase the standard of the project outcome.
Machine learning	Outcome forecasting	• Provide predictions about the project outcome. • Help in risk assessment and contingency planning.
Agile project assistant	Agile methodologies	• Enhance the agility of the project implementation. • Support risk assessment. • Contingency planning
Speech recognition software	Communication management	• Streamlines communication within teams and with stakeholders. • Enhance clarity and record keeping
Natural language processing (NLP)	Documentation and reporting	• Automates the creation and analysis of project documents and reports. • Saving time and improving accuracy
Decision support systems	Strategic decision making	• Helps project managers make educated decisions by offering data-driven insights.
Robotic Process Automation (RPA)	Routine task automation	• Automates repetitive and routine tasks, freeing up human resources for more complex project issues.

However, difficulties regarding project management can be minimized by using AI technologies (Fotso et al., 2022). Then, it will be very beneficial for project managers to perform their day-to-day tasks efficiently without using traditional methods. AI possesses the capacity to revolutionize project management by boosting success rates and productivity (Dam et al., 2019). It will overhaul the practice of project management, automating routine responsibilities like delivering risk estimates and predictions, expediting project planning, and making practical recommendations. Algorithms of AI are capable of analyzing enormous volumes of data from diverse origins like social network sites, project management software, and other databases (Nagireddy, 2023). Because of how quickly modern technology is developing, AI is imperative for project management, and complex projects will not be able to be handled easily by using previous techniques (Mahmood et al., 2023). AI can support the creation of documents such as work breakdown structure, Gantt charts, network diagrams, or risk analysis and can forecast time duration, costs, and potential risks, so AI is simply changing the project management processes in companies (Ledesma, 2014).

Bordea et al. (2020) article examined respondents' perceptions of the anticipated advantages of AI adoption in PM (Figure 4) across various organizational positions, as shown in the figure below. Agile leaders/PMO directors report the most significant impact of AI on productivity (23%) compared to other jobs, implying that AI technologies are incredibly successful at increasing productivity in agile organizations. IT/IS workers perceive the most significant influence on decision-making (20%), which might be attributed to AI's role in delivering data-driven insights critical to IT choices. Executives/functional managers report a significant influence on productivity and decision-making (20% and 19%, respectively), highlighting AI's strategic relevance at senior management levels. Project team members and management have equal impacts across all areas, indicating a comprehensive integration of AI into their everyday operations. (figure 4) This graphic demonstrates how AI may be adapted to specific positions within an organization, enhancing the skills of different departments (Bodea et al., 2020).

Figure 4 - Expected benefits of AI adoption in PM



Historical project management data is unambiguous and less organized. So, Auth and his team created an AI conceptual framework for the project management industry by combining AI development and its ability to automate project management by categorizing some project management applications, such as AI-powered PM solutions and project management bots (Auth et al., 2019b). Vargas and Viana (2015) attempt to forecast the expenses for project management in a project (Vargas & Viana, 2015). Wang (2019) examines the impact of artificial intelligence on project management by highlighting the need to comprehend and utilize AI to boost competitiveness properly. He suggested several AI tools that project managers may use by using a qualitative approach, such as robotic assistance, machine learning-based project management, interconnection and digitization, and autonomous project management. These options are ranked using the weighted scores obtained from the MADM approach. Bhavsar, Shah, and Gopalan (2019) have focused on artificial intelligence strategies that are effective in project estimating and project identification of essential variables.

Bodea, Mitea, and Stanciu (2020) have determined the features of AI adoption in project management by referencing theoretic frameworks associated with technology adaptation in their research. A paper by Fotso, Pradhan, and Sukdeo (2022) discussed AI’s significance for the technological management of projects and specifies the necessary conditions for effective project management as senior project managers, milestones in project management, administration, and budgeting. An evolutionary project success prediction model, or EPSM, has been proposed by Ko and Cheng (2007) in their research on PM in the building sector, and the development of the model is predicated on a hybrid methodology that blends algorithms based on genetics, reasoning with fuzzy logic, and neural network. The study found eight critical elements related to management. Mahmood et al. (2023) explain the applications of AI in project management and how it may benefit to enhance projects across a variety of broad and varied domains (Mahmood et al., 2023).

The article by Dam et al. (2019) proposed a structure for integrating and adapting various AI systems to assist several facets of project management by making use of tasks such as sprint planning, risk management, creating user stories, backlog item identification, and refinement will be simpler to complete with AI tools than they will be with human intervention. The paper by Nagireddy (2023) has studied the ways AI may improve project management performance by outlining the primary challenges teammates have to conquer to accomplish project objectives and the advantages of AI in project management (Nagireddy, 2023). 60% of respondents claimed they conducted the research without using any AI technologies, which demonstrates that project managers included in the study could be underestimating the value of AI systems and failing to see their full potential and applications. Naggiredy (2023) concluded that many people are not aware of the opportunities of AI technologies (Nagireddy, 2023).

The study by Fridgeirsson et al. (2021) investigated the advantages of AI in project management in each of the ten PM knowledge area categories as outlined in the PMBOK. Taboada et al. (2023) investigated the function of artificial intelligence in developing project management, and they argued that the PM community will integrate AI technology into PM methodologies. Raharjo (2023) discussed the difficulties and effects of using AI in project management with an emphasis on how these issues relate to process groups outlined in the PMBOK (Hashfi & Raharjo, 2023). It seeks to advance knowledge of the integration of this topic and offers insights into the difficulties and effects that arise. The article written by Elmas and Babayev (2021) explains the application of AI to PM using some techniques.

Some studies focus on Machine Learning in project management. The paper by Nieto and Vargas (2023) discusses a mobile application that is capable of letting the CEO know about changes that require attention in order of importance, considering the dangers involved, and coming up with solutions for each. It also can suggest which approach the team should take to achieve their goal and create a few algorithms to use ML and AI in PM.

A structure for the autonomous planning of projects with an expert system linked to Revit BIM software was made available by Alijebory and Qaisissam. (Taboada et al., 2023). Mhlari's (2020) research also primarily focuses on artificial intelligence methods that may be used in projects to lower the risk and improve project risk management by monitoring issues and attempting to find out whether any AI solutions were already in place to help with them. The findings suggested that using machine learning apps might help with risk monitoring difficulties. Also, Mhlari's (2020) mentioned the advantages of applying machine learning to the management of risks in a project. Additionally, P. Zhou and EL-Gohary (2015) reported on a highly successful machine learning text categorization system for classifying building components in ecological regulatory papers. Cheng et al. (2020) have developed a model that can forecast the efficiency of a construction model, and it incorporates ML-based least square SVM (M.-Y. Cheng et al., 2020). ML and AI are capable of drastically altering how projects are carried out; AI is not biased towards optimism, and it can extract meaning from complicated data far more quickly and on a much larger scale than people using conventional project management tools. (Taylor, 2021).

To improve several agile project management challenges, Hoa Khanh suggested a framework that combines AI tools, including DL and ML, appropriate for effort estimates (Dam et al., 2019). A brief overview of the research on the application of AI to PM was conducted by Marchinares and Aguilar-Alons (2020), who concluded that the most popular techniques for estimating software effort and forecasting the performance of the project. To assign qualified software developers to a particular project, Javeed et al. (2020) presented a deep learning-based method to assess the software developer's coding proficiency by examining previously produced source code. To enhance the estimation of the effort, planning, and changing requirements in software projects, Nayebi et al. (2018) created a decision support system employing NLP (Natural Language Processing). A framework for an AI-assisted chatbot for project management was presented by Cirule and Berzisa (2019). This approach could cut down on project failures and save valuable time for project managers.

2.4.2 Uses of AI in Project Management

Automation has several definitions, and according to Taylor, the capacity to organize and integrate resources such as people, technologies, and procedures through a predetermined workflow is known as automation. (Taylor, 2021). Currently, project managers do many repetitive tasks manually. If AI can handle them, project managers can put their valuable time into practical and innovative tasks such as network development, people management, project vision, and team building (Aghlamazyan, 2022). AI can monitor the project team members, work automatically, keep tabs on their progress, and also notify the project manager when action is needed (Lahmann, 2018). They may find relief from some duties, such as data input and administration, project plan development, and more, using AI assistance (Elmas & Babayev, 2021).

With the use of algorithms, AI can help project managers assign tasks to specific teammates based on their skill sets (Nagireddy, 2023). These algorithms can evaluate the abilities of each team member, their training, and historical performance, and they can decide which member is most qualified to do particular tasks. AI can support planning the sprint of the agile project environment and is capable of evaluating the complexity of the upcoming sprint and anticipating how long activities will take (Ledesma, 2014). Additionally, it can indicate areas where team members require extra training, making project management easier (Nagireddy, 2023). Managers might teach weak members before starting a project to improve their skills. The manager can ask questions such as which projects carry the highest risk. What is the project that I need to worry most about? Finding answers to these types of problems that managers can solve quickly without going through dozens of lists and spreadsheets (Aghlamazyan, 2022). This is about forecasting the project's future, such as managing possible risks throughout the project and managing costs (Aghlamazyan, 2022). The paper written by Lahmann (2018) mentioned that Artificial Intelligence may soon be able to transform mental maps made by projects. One of the main worries a project manager has when their projects are directly competing with other projects is that they require the same skill set. Still, the PM can measure the effects of resource disputes by doing a “cause and effect” analysis using AI (Taylor, 2021).

AI can be used for strategic partnerships because it will compel managers to collaborate with people who share goals for expansion (Aghlamazyan, 2022). Improving team communication is another important application of AI for project management. Team members of a particular project may exchange information, get real-time feedback, and connect by using some communication tools (Teams, Zoom, and Google Meet), which help prevent misunderstandings and mistakes and expedite the decision-making process (Nagireddy, 2023).

Chatbot robots that can act as project assistants and primarily rely on speech or text recognition bots will contribute to interactions between humans and computers (Lahmann, 2018). It is a software program that uses text or text-to-speech to carry out text-based or live agent conversational instead of requiring direct human interaction, and it can communicate with individuals via messaging apps and automate talks (Taylor, 2021). Chatbots can use natural language to schedule phone meetings with human representatives automatically. The individuals who are talking on the phone do not recognize that they are conversing with a machine (Auth et al., 2019b). Johnk et al. (2020) have mentioned several types of bots: project assistant bot Stratejs, which the project teams can receive assistance from the bot by sharing daily duties, generating new tasks, or asking colleagues to complete unfinished business; red booth bot, which the team may see the issues and daily tasks that they are working on, a virtual assistant called PMO to which uses machine learning to make recommendations about the cost, time duration and resources about a particular task, AI-based platform Cloverleaf which facilitates learning about the working methods of the project team members (Johnk et al., 2020).

The phrase “Project Management Bots (PMB)” was introduced by Gartner in 2017 in the Hype Cycle for Project and Portfolio Management report (Auth et al., 2019b). PMBs are inclined to have interfaces for writing or speaking designed for interacting with humans, resembling the characteristics of Chatbots. Robots that have AI technology can perform various tasks. AI bots are capable of completing fewer complex functions by cutting down 50% of the project manager's duties (Victor, 2023). Time may be saved, and project managers can focus more on critical objectives, employees, and other processes that support their strategic management.

According to past project information, AI may provide enhanced procedures that facilitate project operations and increase efficiency (Nagireddy, 2023). When required, AI facilitates the gathering of detailed information, the assessment of the project’s possible risks, and the creation of a warning (Elmas & Babayev, 2021). Computer systems are known as artificial neural networks (ANN). Neural networks (NN) are modeled after the biological NNs found in human brains that are built on nodes that stimulate actual brain neurons (Taboada et al., 2023). Synapses link these nodes and facilitate information transmission (Taboada et al., 2023). It ensures that the undertaking is carried out by the parameters and plan determined throughout the preparation process stage (Elmas & Babayev, 2021). Processes will move more quickly when AI is applied to human resources. AI uses historical data to learn and anticipate project costs with accuracy and AI can lower personal costs without sacrificing the output quality. An ideal project timeline can be produced using AI, and an updated timetable can be instantaneously delivered if the project’s scope or resource allocation changes. ML allows businesses to succeed and return on investment, as well as an improved perspective of risk.

Inside the business and an improved balance in the project portfolio (Nieto-Rodriguez & Vargas, 2023). Another capability is that it can eliminate human prejudice from decision-making and monitor the project's success. Machine learning enables forecasting and provides advice (Lahmann, 2018). Supervised learning, unsupervised learning, reinforcement learning, and rules-based learning are the four strands of machine learning (Taylor, 2021).

2.4.3 Challenges of AI Use in Project Management

Depending excessively on AI has the potential to weaken a person’s critical thinking. It is a set of skills that are necessary for tackling challenging issues and coming up with novel solutions (Nagireddy, 2023). Usually, a person has those skills from birth. To use AI effectively in project management, an elevated degree of maturity and a large amount of data may be needed (Mahmood et al., 2023). Even though AI is beneficial to all project managers, AI devices cannot be as creative or emotionally intelligent as humans (Nagireddy, 2023). Another challenge of using AI in the project management area is that it can mimic human cognitive processes, including problem-solving and decision-making (Hashfi & Raharjo, 2023).

According to the report of Wang (2019), three primary worries prevent AI from being widely utilized in project management, including confidentiality problems in AI-driven advertising, fears of losing one’s job, and fears of complicated AI systems leading to failure and the lack of knowledge of using AI for project management. According to his study, 87% of participants stated that AI will alter their line of work over the next three years. Accuracy is another challenge. For systems that use AI and ML, it has been noted that although programs might be able to mimic human behavior, practical applications of AI are unable to employ human techniques because accuracy cannot be guaranteed (Crawford et al., 2023).

One study has found that the contributors of the study believe AI will primarily impact project cost management, and it is nearly 58%; 1% of respondents said AI will have an excessive effect on schedule management and 47% on risk management. Also, AI would have a medium-high effect on (61%) project procurement management (Fridgeirsson et al., 2021). So, these are some challenges that can be encountered when adapting AI technologies.

2.4.4 Impact of AI Integration on Project Agility

The acceptance of project managers to make alterations in an ongoing project to attain the desired project outcome is known as the agility of project management (Shafiabady et al., 2023). Agility in project management relates to the ability to rapidly adjust project processes and deliverables in response to changing circumstances, thereby maximizing value for stakeholders (Highsmith, 2001). Nowadays, many organizations use novel technologies. AI is gradually becoming integrated into project management to improve adaptability while paving the way for organizations to adapt according to the changes needed by the client (Tominc et al., 2023). This study is mainly focusing on the integration of AI within project management.

Several studies have investigated the impact of AI on project agility by explaining its advantages and drawbacks (Tominc et al., 2023). Wamba et al. (2020) highlighted that organizations can improve responsiveness by using AI to make predictions about the reaction to fluctuating project needs and market circumstances. After evaluating previous information and identifying new patterns, using AI provides the capability for project managers to identify possible hazards and take necessary actions to prevent those (Shamim & Islam, 2024).

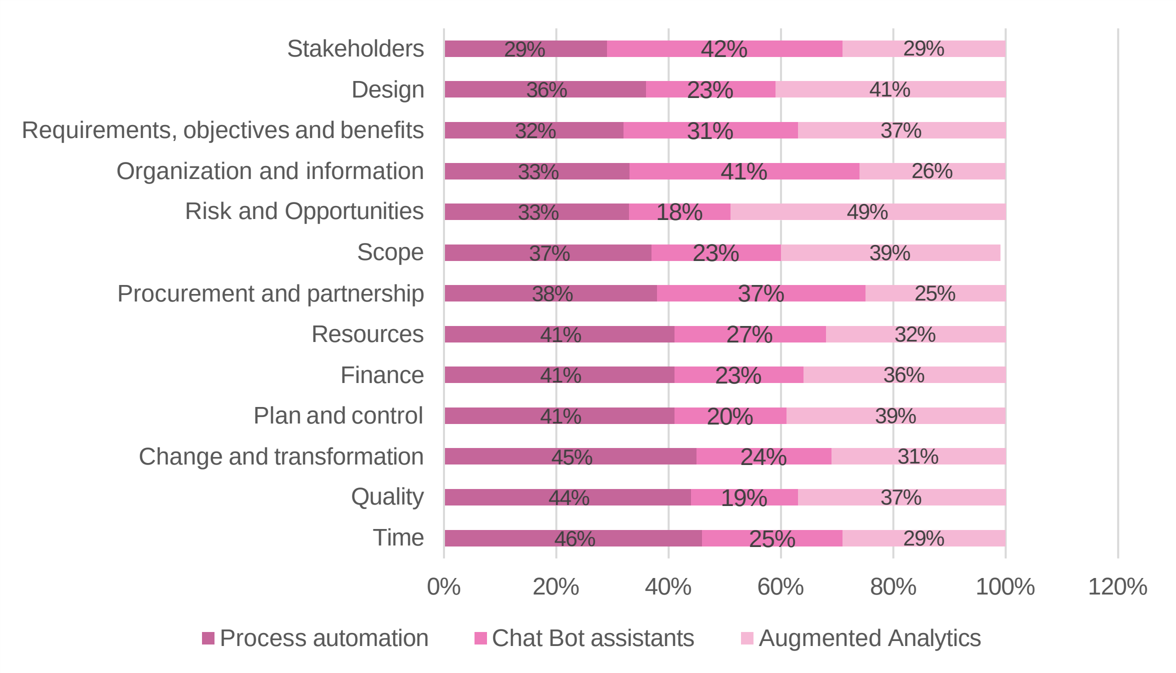
Wamba et al. (2020) discovered that utilizing AI technologies will enhance the flexibility of the project (Wamba-Taguimdje et al., 2020). The use of AI-driven tools to allocate resources among team members within the project team, planning, and risk administration improve the ability of project managers to adapt according to the relevant changes within the project time (Bainey, 2024). Insights based on AI gave project managers real-time visibility into project success indicators, allowing them to make data-driven choices and modify project tactics as necessary. Tadimarri (2024) study discusses that when compared to the organizations that use traditional project management methods, one who uses AI is delivering the projects faster. By automating mundane operations, optimizing workflows, and promoting more outstanding communication and cooperation among project stakeholders, AI technology helped project teams optimize procedures and minimize project deliverable time to market (Tadimarri, 2024a).

While empirical data indicates that AI integration might improve project agility, various problems and issues must be addressed. To increase the effectiveness of the utilization of AI, it is essential to have a highly standardized data set which needs to be very relevant to the study area. There is no point in using inaccurate data for AI predictions because it may lead to faults in the outcomes. Also, organizations should make sure that the data they are using are accurate, complete, and accessible to generate successful insights and predictions from AI algorithms (Zha et al., 2023). That is the main thing that organizations need to make sure of.

AI adoption necessitates organizational and cultural adjustments. To properly utilize the possibilities of AI-driven tools, project teams must be taught AI technology, and organizational procedures may need to be modified (Rajagopal et al., 2022). Utilization of AI may lead to some ethical and legal problems, mainly with the protection of data, bias, and openness. Organizations must adopt ethical principles and governance frameworks to support responsible AI deployment and risk mitigation (Patel, 2024).

The below figure shows the results of the research done by Bodea et al. (2020).

Figure 5 - The expected impact of AI adoption on PM



Empirical research discusses the utilization of AI within project agility, such as responsiveness, flexibility, and delivery speed. Companies can use this AI to improve the decision-making process and improve the way of allocating resources (Tadimarri, 2024b). AI technology is an integral part of project management, and the understanding of this technology is crucial in projects like data quality, change management, and ethical issues. Nonetheless, more long-term research has to be carried out in order to verify the future implications of AI technology on project agility and organizational performance improvement.

2.5 Organizational Learning Culture

The learning-focused organizational culture of AI that is incorporated into managerial processes is a critical asset that ensures successful AI applications. The literature review explores practical evidence on how organizations are implementing AI technology, creating an environment that encourages innovation and creativity, and using the data generated by AI to improve the agility of projects. Learning organizational theory, presented by Argyris and Schön (1978) and developed by Senge (1990), allows analyzing the way of information processing that helps those organizations to increase performance efficiency and quickly react to changes (Argyris & Schön, 1978; Senge, 1990). Organizational learning theory provides probable answers to many questions that arise during AI integration. Learning in the organization, which Argyris and Shon (1978) first presented and later Senge (1990) developed, includes the development, understanding, and application of skills with the ultimate aim of reaching exceptional performance and transformation. Organizational theory of learning gives significant perspectives on many crucial aspects of integrating AI into project management.

AI technologies permit businesses to get data from different sources like project experience, external data bases, or expertise systems (Senge, 1990).

Ensuring having colossal data sets is a mandatory condition to join AI projects. Liu et al. (2018) discovered that the powered AI analytics applications can extract knowledge and trends from the data sets, which are immense, thus giving the project teams valuable information that can be used for decision-making and strategy formulation (Liu et al., 2018). The articles consider the role of communication tools and collaboration infrastructure with regard to AI-based information dissemination, cross-functional cooperation, and organizational learning. Organizational learning theory is based on the idea that organizations are able to use information in accordance with what makes things better and at the same time, they are adapting to the changes. This theory stresses that the meaning and the way of learning are both critical (Argyris & Schön, 1978). Project teams can face, resolve, and later modify ongoing needs by using AI tools (Li et al., 2024c). Thereby AI will support project managers to achieve a successful outcome at the end of the project. By encouraging critical thinking and reflection, organizations can capitalize on AI-driven insights to enhance project agility and performance.

2.5.1 Organizational Learning Culture and Project Management Agility

Organizational learning culture and project management agility are two interrelated ideas that are very important for the success of projects in organizations. Organizational learning culture is about the group norms, values, and practices shared across the organization, which function as critical drivers for learning, sharing of knowledge, as well as innovation (Khan & Saengon, 2021). This culture encourages the employees to be self-conscious, to try out new ideas, to test different approaches, and to reflect on their previous experiences in order to enhance their individual and collective performance (Ivaldi et al., 2022). Further, they have indicated that the organizational learning culture of the teams will cope with the difficulty of changing situations, application of new technologies, and implementation of a continuous process of improvement.

Psychological statistics show that project performance has many potential outcomes when the organizational learning culture is present. For instance, Nachbagauer (2022) states that companies with a strong learning culture usually display a higher degree of flexibility and resilience in their project management practices. Unlike a single team that struggles to respond to changes in project requirements, these organizations are more flexible to react to changing conditions; thus, the outcome is more robust as compared to individual teams. Rehman & Sohaib (2023) observed that the crucial role of enhancing each other in a project through facilitative environments was emphasized, and resultantly, the members of the team were able to capitalize on their collective expertise and insight in order to overcome challenges and achieve the project objectives.

Not only does the organizational learning culture enhance agility, but it is also related to increased innovation and competitiveness. Organizations can stimulate their employees to go beyond the margin when proposing new ideas as well as making experiments with various approaches, thereby creating an atmosphere of innovations that drives continuous improvements for the sustainability of growth (Khan & Saengon, 2021). Also, learning organizations can quickly adapt to external changes, and they have the potential to make the most of emerging opportunities. This puts them in a competitive position as compared to other organizations. Despite the fact that the advantages of the organizational learning culture are widely known, the process of its creation is quite tricky. The establishment of a conducive work environment by way of strong leadership support and relentless commitment to improvement at all levels in the organization (Ausat & Shafiq, 2024).

In addition, organizational learning is an ongoing practice that requires venture and feeding over time. Therefore, the organizational learning culture is the key to project management agility and organizational performance improvement. Such organizations will experience fewer distractions and, therefore, will be better suited to adjusting to any changes that occur as well as to make inevitable improvements while staying ahead of their competitors. Nevertheless, the implementation and maintenance of an enduring learning-oriented environment is solely dependent on leadership, knowledge management systems, and a deep-rooted willingness for innovation.

2.6 Change Management Process

Change management can be considered as one of the processes employed by organizations where planning, implementation, and control of changes to ease the impact of the transitions are done, and the resistance from the stakeholders is minimized (Ayele, 2020). The literature on change management highlights its role in the process of organizational adaptation and the importance of strategic initiatives in making an organization move forward. Diverse change models and frameworks are the other vital things to consider as the change management process goes on. Taking as hands-on, Lewin's three-stage model of change, which comprises unfreezing, changing, and refreezing, proposes how organizations can gradually overcome the various transitions (Qwesha, 2019). In addition to this, the eight-step model of Kotter (1996) also highlights the significance of creating a sense of urgency, building a guiding coalition, and enabling the employees to drive the change. While they do not give universal answers, these models provide priceless information about how the change process should be managed for organizing more structured transformations.

Moreover, along with the understanding of change models, the leadership part in steering change management is also essential. Such change leaders can visualize, communicate, and be emotionally intelligent to inspire and motivate employees throughout the change process. Leadership support is essential in this endeavor as it can remove individuals' suspicion towards change, and it enables everybody in the organization to work for a common aim.

Furthermore, communication is a significant component in change management, allowing for the formulation of general perceptions, solutions to affected challenges, and motivation of all stakeholders (Naeem, 2020). The openness and transparency of communication channels allow the employees to understand the logic of the changes, to express their opinions, and to bring their contribution to the change process (Kotter, 1996). As well as that, communication helps you arrange the expectations and relieve uncertainty; thus, the adaptation process is smoother, and the rate of employee satisfaction is higher.

2.6.1 Change Management Process and Project Management Agility

The literature points the way to positive connections between the processes involved in organizational change and project management's agility. Change management practices help the implementation of agile methodologies in project management, and that is why organizations can now respond quickly and effectively to changing requirements and market conditions (Naeem, 2020). Effective change management provides teams with all the tools, knowledge, and support structures needed for the change process. This helps teams to be more agile and adaptable, using iterative development, continuous feedback, and collaboration (Arefazar et al., 2022). Through the inclusion of change management processes in the list of methodologies that are used in project management, the entity, therefore, becomes more adaptable, improves the outcomes of the projects, and readily adapts to the rapidly changing business needs and customer preferences.

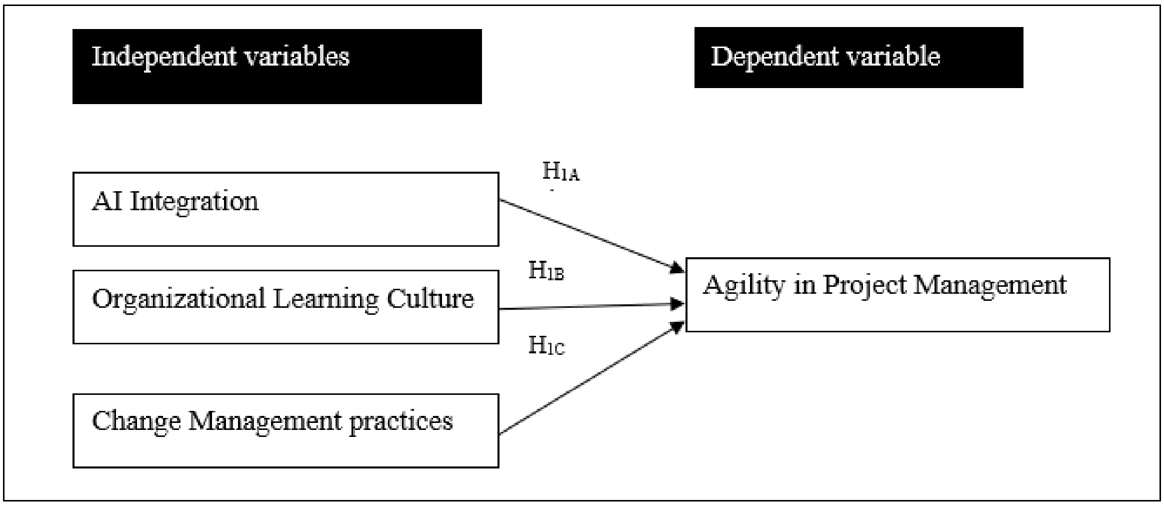
3 Research Methodology

3.1 Introduction

This chapter describes the study methods used to determine the influence of AI integration on project management agility. The methodology refers to the conceptual framework, research design, data gathering methods, and data analysis methodologies used in the study. The results of the research were acquired from both primary and secondary sources to emphasize the findings of the research article, using a mixed method strategy that included a survey and an interview with project managers working in Norwegian organizations. Therefore, this decision was made in order to offer a better understanding of the success of AI in project management in practice, as well as to propose techniques for its better use.

3.2 Conceptual Framework

Figure 6 - Conceptual Framework



3.3 Application of the Conceptual Framework

The conceptual framework for this study consists of three independent variables and one dependent variable, as follows:

AI integration (Independent variable)

This element evaluates how AI technology is adopted in workflow processes and management methods (Patel, 2024). It refers to the use of AI-powered technologies, the amount of AI knowledge in the organization, and the inclusion of AI in project workflows.

Organizational learning culture (independent variable)

This characteristic is that the company has a culture of learning that involves talking about essential things and also training each other (Senge, 1990). This applies to elements such as the leaders' support of the learning initiatives, the employee training and development programs, and the promotion of knowledge sharing and collaboration.

Change management practices (Independent variable)

This is the variable representing the management of change and the obstacles of resistance to AI integration (Venkatesh et al., 2003). It comprises techniques like awareness of stakeholders, communication plans, training programs, and incentives for accepting AI technologies.

Agility in Project Management (Dependent variable)

Agility variables thus represent the general effectiveness of an organization in handling project management processes (Satinder & Venugopalan; Navaneethan, 2015). The concept comprises features like the capability to adjust to change, adaptability to changing needs, and project speed, which are the factors that lead to better project results and stakeholder satisfaction.

AI integration, Organizational learning culture, and Change management practices together affect Project agility by working through the processes of knowledge management. Companies can better react to the frequently changing needs of projects and the environment by shifting the organization's mood to a continuous improvement principle, adopting quality change management methodologies, and introducing AI technology into project workflow.

3.4 Hypothesis Formulation

Based on the conceptual framework outlined in Section 3.2, the following hypotheses are formulated.

H1: There is an impact of AI integration on project management agility in Norwegian companies.

H2: There is an impact of organizational learning culture on project management agility in Norwegian companies.

H3: There is an impact of change management practices on project management agility in Norwegian companies.

These hypotheses propose specific relationships between the independent variables (AI integration, Organizational learning culture, and Change management practices) and the dependent variable (Project management agility).

3.5 Research Design

Mixed methods approach

This study uses a mixed-method approach to explore project managers' perspectives towards project management agility in the purview of AI integration, Organizational learning culture, and Change management process. The strategy, which consists of both qualitative and quantitative approaches, is well-designed and well-equipped to render in-depth and broad-ranging analyses of research topics and concepts. At present, the research is being carried out in a mixed-method approach which combines quantitative and qualitative methods. The mixed method is the better research technique for the kind of question and also the study subjects.

Quantitative approaches based on qualitative data are applied when data is qualifiable, for instance, numbers and scales. In this case, quantitative research is the data collecting for project management agility. This second step involves exploring, analyzing, and studying the data gathered from the survey designed by the researcher. The quantitative method consists of carrying out statistical analysis of numbers in order to identify correlations among the variables (Bryman, Bell, 2015, 012). With this approach, measuring variables such as AI integration, Organizational learning culture, Change management practices, and Project management agility through the use of structured survey instruments is possible.

This is the other part of the research method, which was the qualitative method, that was done through semi-structured interviews. These interviews were conducted in order to gather insights regarding personal experiences and views on AI integration to improve the project management of the company. After the questionnaire had been filled in, the interview with the project managers began. Apart from quantitative research, qualitative research may also be used to examine potential constraints and problems regarding AI adoption for effective project management.

The mixed method approach may assess the advantages and drawbacks of employing AI for managing projects. Combining both quantitative and qualitative methods can provide a better knowledge of the issue and indicate areas for further research and development. This process assists researchers in evaluating their findings, ensuring complementary outcomes, and explaining unanticipated (Dawadi et al., 2021).

3.6 Population and Sampling Method

3.6.1 Population

The population of the research is the group of people, objects, or events that the researcher wishes to make conclusions (Gay & Airasian, 1999). This study's demographic consists of project management experts from a variety of sectors including managers of projects, teammates, and stakeholders employed by Norwegian organizations. These individuals are in charge of designing, implementing, and monitoring projects for their respective organizations.

Project managers come from industries such as finance, IT products and services, general business, healthcare and pharmaceuticals, consumer goods and retail (electronic, apparel), manufacturing and industrial (automotive, machinery), telecommunication, etc. These project managers' ages and gender are varied, and it entails the impact of age, gender, and experience in the application of artificial intelligence in directing projects. Also, they are at different management levels, such as higher-level, operational, and lower-level management. Therefore, the degree of technological aptitude and uptake within an organization may vary. This dimension allows for a comparison between adaptation to AI in project management and those of others who want to follow the traditional method. Furthermore, the sample includes project managers who are currently using AI, who have never used AI, and people who like to use AI to complete their tasks efficiently.

3.6.2 Sampling Method

A sample can be defined as a group that the researcher used to collect data (Bhandari, 2020a). There are two types of sampling methods: probability and non-probability sampling. This research has used non-probability sampling, which is a method that uses non-random characteristics such as availability, geographical closeness, or expert knowledge of the people.

The researcher likes to interview to address the research problem (Nikolopoulou, 2022). The sampling approach which was used in this study is the convenience sampling technique because it allows gathering required data from the people who can conveniently participate in the survey and the interview (Nikolopoulou, 2022).

The sample for this study consists of project managers who work in Norwegian companies. A subset of respondents selected for interviews from the sample used in the quantitative phase. The project managers who live in Norway have been chosen as the sample on LinkedIn, and the survey was sent by email.

3.6.3 Sample Size and Determination

The number of participants who are involved in the survey and the interview is called the sample size (McCombes, 2019). The population sample and sampling method outlined above provide a systematic approach to selecting representative samples of project management professionals and collecting data to address the research objectives.

The sample size needs to be more prominent than 30, and lower than 500 is sufficient to achieve more accurate data (Serdar et al., 2021). The current research sample size is 84 project managers, and to obtain a deeper understanding, interviews were conducted with seven project managers from this sample.

3.7 Data Collection Method

Structured questionnaires were distributed to a sample of project management experts to obtain quantitative data. The survey instrument was created to collect data on AI integration, Organizational learning culture, change management techniques, procedures, and Project management agility, according to the conceptual framework given in Chapter 3. To acquire a thorough grasp of the issue, qualitative data was gathered through interviews with project managers who took part in the survey. The questions for the interview were also created considering the variables addressed in Chapter 3.

Primary Data

Primary data are those that the researcher has created directly for the goal of solving the study problem through the investigator's effort and experience (S, 2016) and are also referred to as first-hand or unprocessed data. Data was collected accurately and methodically within the scope.

In this research. To provide a valid and thorough dataset, mixed methods were adopted, including both quantitative and qualitative techniques.

For the quantitative phase, a structured survey will be distributed electronically to a sample of project managers who work in Norwegian companies to examine their thoughts and opinions about the subject area. Furthermore, respondents are being allowed to describe specific project management tasks that they believe might benefit from AI integration, as well as the type of training they believe is necessary to employ AI within project management effectively. This survey was meant to capture qualitative as well as quantitative responses, giving a basis.

For the qualitative phase, semi-structured interviews have been conducted with seven chosen project managers to get a deeper understanding of the subject area.

Secondary Data

Secondary data is data that has been acquired and documented for reasons unrelated to the present research topic by someone other than the user (S, 2016). It is the easily accessible form of data gathered from several sources, including websites, books, journal articles, government publications, internal organization records, and reports (S, 2016). Secondary data was acquired by referring to existing journal articles, books, and surveys that had been previously performed. Information from these was used to formulate the questionnaire and interview guide.

Survey Instrument Development

Surveys allow for the collection of data from a large sample of respondents in a structured and standardized manner (Bryman & Bell, 2015). By administering surveys to project managers, team members, and other stakeholders in various organizations, this study can gather quantitative data on the variables of interest and test the proposed hypotheses. The questionnaire is prepared depending on the study questions and hypotheses provided. The questions were created to look at the current status of AI implementation in project management. Participants were asked to respond using the Likert scale, and there were some open-ended questions as well. The Likert scale results are used to determine how much these project managers know about artificial intelligence (AI) techniques, how satisfied they are with their work, whether or not they use AI, and what proportion of people use AI for project management tasks. Also, I referred to the book written by Peter Taylor to get the required knowledge for creating this questionnaire (Taylor, 2021).

First, there are some questions to get to know about the background of the person who answered the questionnaire. There, I asked about the gender, age, the level of the role in his or her company, and the primary source of revenue of the company. These questions were used to analyze what level of managers use AI most and what industries use AI solutions for project administration (Fridgeirsson et al., 2021). Another question is designed to measure the

Participant's familiarity with the industry of project management. Also, there are some questions to get to know about the awareness of AI-based systems in the company.

Furthermore, additional questions were designed to assess the participant's understanding of how much they use tools, whether they are helpful to project managers, whether they are satisfied with the current process with or without the use of AI, whether AI will assist in identifying areas that require training, and what they think about the impact of AI within project management.

Interview Protocol Development

An interview procedure was created to help lead the semi-structured interviews and it comprises a series of open-ended questions on the study objectives and hypotheses. Semi-structured interviews were conducted with selected respondents to gain deeper insights into the context and complexities that the survey data may reveal.

The qualitative phase entails conducting interviews that are semi-structured with selected project managers who are experts in extensive knowledge and practical experience to gain a greater awareness of how much the companies use AI for project management (George, 2021) and to learn about their perspectives and ideas on the possibilities for applying AI in project management, as well as the specific activities they believe may benefit from AI integration. Interviews comprise a series of open-ended questions on the study objectives and hypotheses. Those were performed using Teams software (Audio Conversations) according to the preferences of the participants. The people who responded to the researcher’s email after completing the survey were invited for an interview.

3.8 Data Analysis

The final data was analyzed based on the results from the survey and from the interviews that were conducted with randomly selected project managers. Quantitative data has been analyzed using SPSS. Thematic analysis is conducted to analyze qualitative data. The statistical analysis for this study will involve the use of several tools and techniques to test the hypotheses and analyze the relationships between variables.

Frequency Analysis

Frequency analysis refers to the pattern of frequencies in each variable, which is about how many times each potential value appears in a dataset (Turney, 2022). Frequency analysis was done in this study to explore the responses to the questionnaire, and it was performed for each categorization. Each classification variable is subjected to frequency analysis.

The frequency distribution, which is stated as a percentage, seeks to tally the number of replies associated with different values of a particular variable. From the result, we can gain a better understanding of the respondents' backgrounds, such as gender, age group, working sector, and feelings about AI.

Reliability Analysis

Reliability analyses are usually based on a combination of multiple response measures, which is a very typical approach in research. A multi-dimensional scale has been developed to recognize and model the structure and dimensions of the dataset (George & Mallery, 2018). The reliability of the measurement is tested based on the strength of stability and consistency (Kimberlin & Winterstein, 2008). Consistency is all about demonstrating the degree to which the overall pool of conceptual data was valid and reliable. Cronbach's Alpha is a reliable measure that shows how tightly the components of a unit are positively correlated (Vaske et al., 2017). It was estimated by using the relationship of the averaged inter-correlations of those items that were employed to measure the sense of it. An increased dependence on inveterate consistency is observed with the approach to one of Cronbach's Alphas as a coefficient value. A reliability test score of less than 0.60 is regarded as bad, and values over 0.70 are regarded as acceptable. A value over 0.80 is desirable. Cronbach's Alpha is close to 1 if the internal consistency of the model is high (Weir, 2005).

Validity Analysis

Validity is the primary measure of how much a concept, conclusion, or measurement is assessed in the main and how much it fits the real world. The term "valid" is derived from the Latin word validus, which means strong (Kumar, 2020). The validity of a measurement tool (for instance, a test in education) is the extent of the tool's capability to measure what it claims to measure. KMO Bartlett's Test is utilized to measure validity. The measuring instruments are accepted when KMO Bartlett's test value is higher than 0.05 (Soulis & Kyriakopoulou, 2024).

Normality

Normality analysis is the main thing to check if data follow a normal distribution, which is one of the most critical assumptions in many statistical tests (Knief & Forstmeier, 2021). In this case, a normal distribution means that the data points are symmetrically distributed around the mean, with the most being close to the mean and the least being on the tails. The alterations from normality could distort the reliability of the statistical tests, which, in turn, could result in biased results or wrong conclusions. Thus, normality analysis gives the researchers a chance to find out whether their data meets the null hypothesis. Some of the methods that are used for normality testing are, for instance, testing skewness and Kurtosis.

Descriptive Analysis

Descriptive analysis was conducted to summarize and organize the properties of a dataset. A data set is a grouping of answers or observations from a sample or the complete population (Bhandari, 2020b). Frequently involve producing tables of quantiles and means, techniques of dispersion such as variance or standard deviation, and cross-tabulations or "crosstabs" that may be used to carry out numerous distinct hypotheses. These theories frequently emphasize disparities between subgroups (Rawat, 2021). Descriptive analysis was performed on the interval-scaled independent and dependent variables, including means and standard deviations. Using this method, the researcher can learn if respondents answer more favorably or negatively, as well as the pattern of their responses.

Regression Analysis

Regression analysis can be employed to evaluate hypotheses (H1, H2, and H3) and determine how effectively other variables (such as AI integration, Organizational learning culture, and Change management practices) predicted the dependent variable's performance. Regression analysis explains the impact each independent variable has on the dependent variable by fitting a line to the observed data.

Thematic Analysis

Thematic analysis is a method for examining qualitative data. It typically refers to a collection of texts, such as an interview or transcripts. The researcher attentively studies the data to uncover common themes, which are subjects, concepts, and patterns of meaning that appear again (Caulfield, 2019). Thematic analysis was conducted to analyze the gathered data through conducting interviews.

3.8.1 Data Analysis Software

Statistical analysis was performed using SPSS (Statistical Package for the Social Sciences) and Microsoft Excel software and systems. Statistical analysis is a vast field that includes descriptive statistics and regression analysis. It is a diverse set of statistical functions and methodologies for data analysis. A thematic analysis that comes out of the qualitative data and the main patterns and themes that emerge. The data collection method employed by the study is a blended method that goes beyond hypothesis testing with quantitative data collection, explores in-depth insights, and validates findings through qualitative analysis, which makes it possible to develop a complete understanding of the consequences of AI integration on project management. This is the most important thing to learn for taking the action and

Proposals for the successful management of the project in the case of AI and organizational changes.

4 Analysis

4.1 Introduction

This chapter contains the final output of the study on the impact of AI integration, Organizational learning culture, and Change management process on Project management agility within the context of Norway's project management field. The chapter starts with the demographic data and frequency analysis. After that, descriptive analysis, reliability and validity analysis, normality test, correlation analysis, and regression analysis were executed. A mathematical analysis has been conducted to describe the qualitative data.

4.2 Quantitative Analysis

4.2.1 Frequency Analysis

Figure 7 - Analysis of Responses

No	Demographic Variable	Details	Frequency	Percentage (%)
1	**Age**	18-30	28	33.33
		31-40	20	23.81
		41-50	20	23.81
		51-60	12	14.29
		Above 60	4	4.76
2	**Gender**	Male	55	65.48
		Female	29	34.52
3	**Sector**	Financial services	11	13.1
		IT products services	26	30.95
		General business	2	2.38
		Consumer goods and retail	2	2.38
		Manufacturing & industrial	12	14.29
		Telecommunication	2	2.38
		Public sector	8	9.52

		Non-commercial organization	6	7.14
		Other	15	17.86
4	**Role**	Higher level manager	10	11.9
		Middle-level manager	42	50
		No managerial responsibilities	32	38.1
5	**Experience**	0-2	29	34.52
		03-May	7	8.33
		05-Oct	22	26.19
		More than 10	26	30.95
6	**Usage of AI in the company**	Yes	33	39.29
		Not yet (Future plan)	32	38.1
		I do not know	9	10.71
		No	10	11.9
7	**Knowledge of AI in project management**	Very good	3	3.57
		Pretty good	16	19.05
		Aware of opportunities	36	42.86
		Limited understanding	25	29.76
		Non-Existent	4	4.76
8	**Feeling about AI in project management**	Scared	0	0
		Concerned	8	9.52
		Intrigued	31	36.9
		Excited	41	48.81
		Not interested	4	4.76
9	**Will the use of AI affect the project manager's role**	Not at all	2	2.38
		Probably	25	29.76
		Possibly	34	40.48
		Definitely	13	15.48
		Do not know	10	11.9
10	**The extent of AI's contribution to the effectiveness of project management**	1	0	0
		2	8	9.52
		3	18	21.43
		4	26	30.95
		5	23	27.38

		6	9	10.71
11	**Usage of AI-based tools to manage projects**	Yes	26	30.95
		No	35	41.67
		I do not know	23	27.38
12	**The extent to of employees use AI-based tools to manage projects**	1	30	35.71
		2	11	13.1
		3	17	20.24
		4	24	28.57
		5	2	2.38
		6	0	0
13	**If so, to what extent do you think AI will be able to support the project manager in establishing and managing project teams**	1	8	9.52
		2	12	14.29
		3	17	20.24
		4	29	34.52
		5	18	21.43
		6	0	0
14	**To what extent do you think AI can help project managers assess the performance of team members**	1	4	4.76
		2	11	13.1
		3	21	25
		4	31	36.9
		5	8	9.52
		6	9	10.71
15	**Do you or other employees in the companies use AI-based tools to manage projects?**	Yes	26	30.95
		No	35	41.67
		I do not know	23	27.38
16	**To what extent do you think such tools are helpful?**	1	14	16.67
		2	7	8.33
		3	10	11.9
		4	30	35.71
		5	14	16.67
		6	9	10.71
17	**How Confident are you in your understanding of the**	1	14	16.67
		2	14	16.67
		3	17	20.24

	Potential benefits of AI for project management?	4	20	23.81
		5	14	16.67
		6	5	5.95
18	**To what extent is your organization ready to accept and adopt new technology for project management?**	1	4	4.76
		2	10	11.9
		3	19	22.62
		4	31	36.9
		5	12	14.29
		6	8	9.52
19	**To what extent is the company's management willing to wait several months, up to a year, to start seeing the benefits of using AI?**	1	4	4.76
		2	13	15.48
		3	26	30.95
		4	14	16.67
		5	21	25
		6	6	7.14
20	**Do you already use language models (ChatGPT or similar) in the project management?**	Yes	56	66.67
		No	28	33.33
21	**To what extent do you use any language model (ChatGPT or similar) within project management?**	1	22	26.19
		2	17	20.24
		3	9	10.71
		4	25	29.76
		5	4	4.76
		6	7	8.33
22	**How do you assess the potential of using language models (ChatGPT or similar) within project management?**	1	4	4.76
		2	6	7.14
		3	12	14.29
		4	38	45.24
		5	15	17.86
		6	9	10.71
23	**Do you think that AI-based systems will be able to help project managers to increase efficiency in terms of administration of**	Yes	64	76.19
		No	0	0
		I do not know	20	23.81

	projects (meeting milestones/deadlines)?			
24	**Will AI to assemble teams and delegate tasks and duties among a**	Yes	49	58.33
		No	9	10.71
		I do not know	26	30.95

Figure 8 - Age

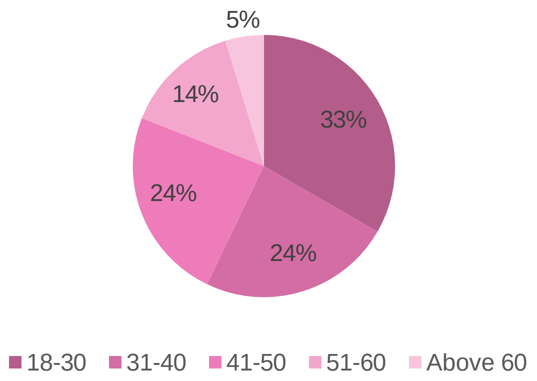


Figure 9 - Gender

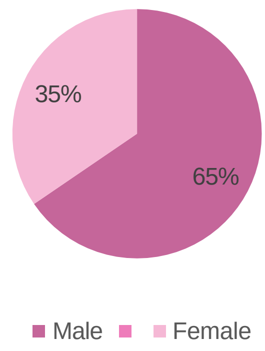


Figure 8 describes the age groups of project managers who participate in the survey, and the highest number of them are between the age of 18-30, which is about 33% of the whole sample; the percentage of respondents between the age of 31-40 and 41-50 is equal. Only 5% are above 60.

Figure 9 indicates that 65% of the respondents are male and 35% female. So, the highest number of respondents who answered the survey were males.

Figure 10 - Industry.

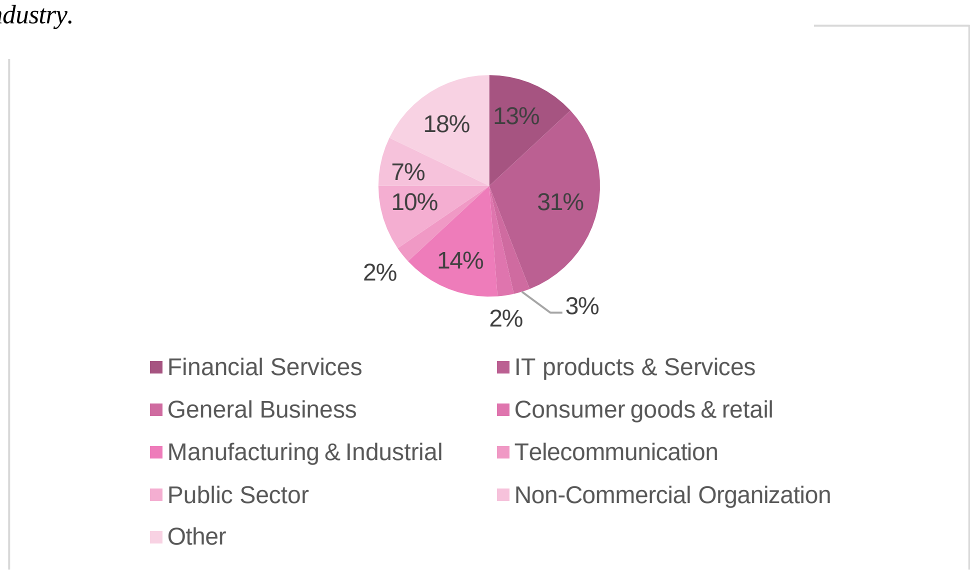


Figure 10 shows which respondents are from which industry. According to this, the highest number of respondents are from IT products and services (31%), while the lowest number of respondents are from consumer goods and retail and telecommunication industries, which is 2%.

Figure 11 - Years of Experience

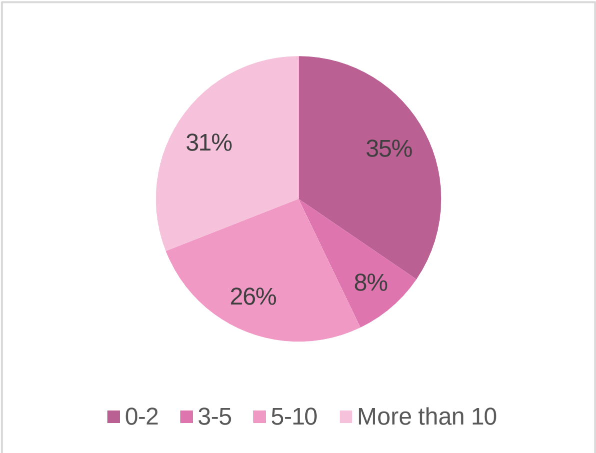
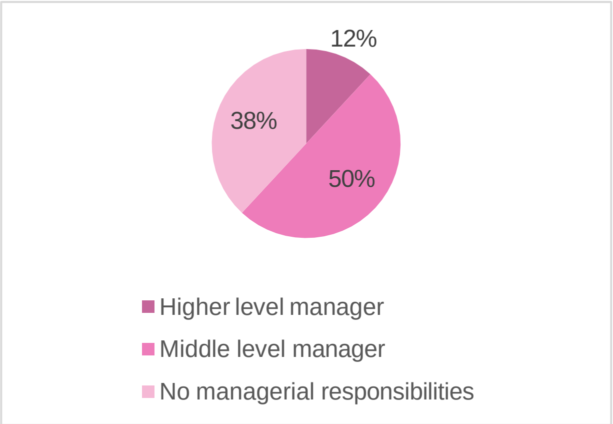


Figure 12 - Role



Of the respondent's project managers, 31% of respondents have more than ten years of experience within the field of project managers. Still, a higher number of respondents have 0-2 years of experience within the field. Only 26% of people have 5-10 years of experience in the field (figure 11).

As shown in Figure 12, the highest number of respondents who participated in the survey play a middle-level management role in their company, which is 50%, and the least number of respondents are from higher-level management (12%).

Figure 13 - Knowledge regarding AI

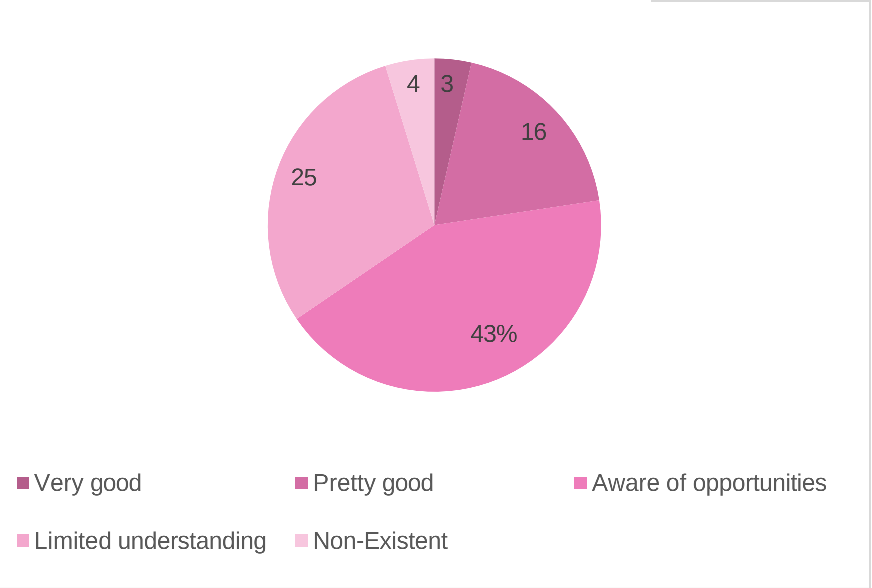


Figure 13 shows the percentage of knowledge regarding AI of project managers in Norway. 43% mentioned they are aware of opportunities of the AI systems, and only 5% responded as non-existence.

Figure 14 – Feeling about AI

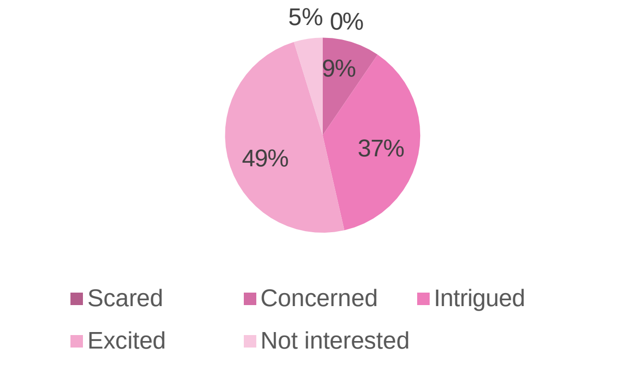


Figure 15 - Effect of AI on project management role

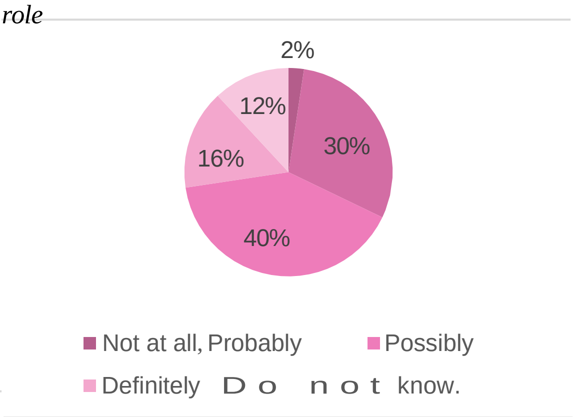


Figure 14 shows the feeling of AI integration into project management within Norwegian companies. 49% are excited about AI, and 37% are intrigued. Only a few percent of respondents have no interest in this (4.8%).

Figure 15 shows the effect of AI on the project management role; 40% of respondents say AI is possible, and 30% say AI will probably affect their role.

Figure 16 - Extent to which AI will contribute to PM

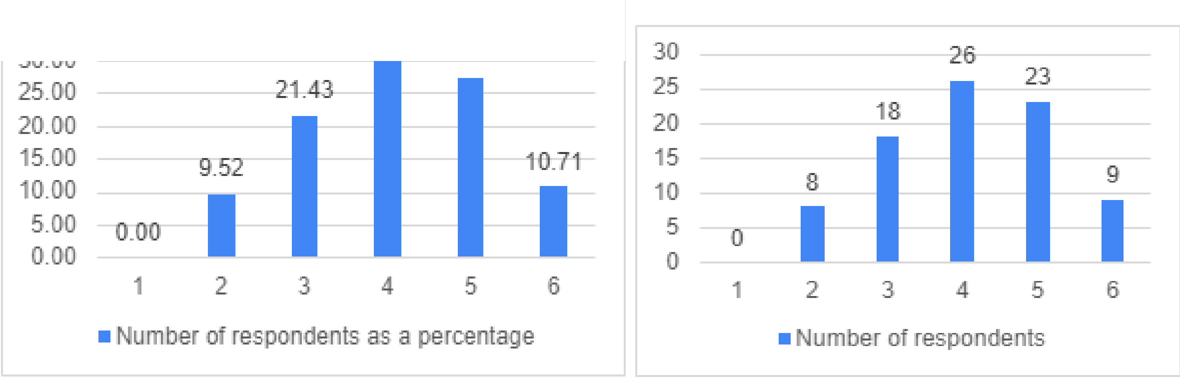


Figure 16 shows the extent to which AI will contribute to more effective project management. A higher number of respondents positively indicates that AI is capable of contributing to making project management more effective.

Figure 17 - Will AI be able to help assemble teams and delegate tasks and duties to each team member

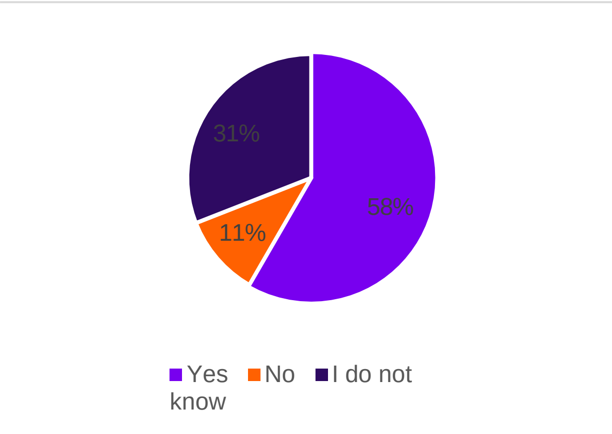


Figure 18 - Use of AI tools to manage projects

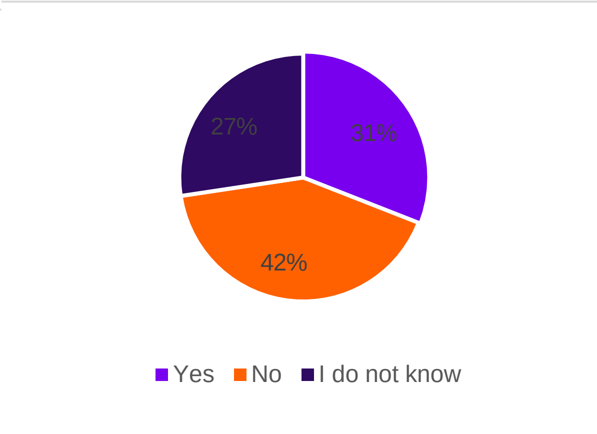


Figure 17 shows respondents' ideas about the help of AI in assembling teams and delegating tasks and duties to each team member. So, 58% responded "yes," and only a lower percentage of respondents said "NO" (11%), while 31% did not know about this.

Figure 18 shows the usage of AI-based tools to manage projects by the respondents or employees working with them. A higher number of respondents said that they do not use AI-based tools (42%). Only 31% mentioned that they use any AI-based tools for managing projects.

Figure 19 - Support of AI for project managers to establish project teams

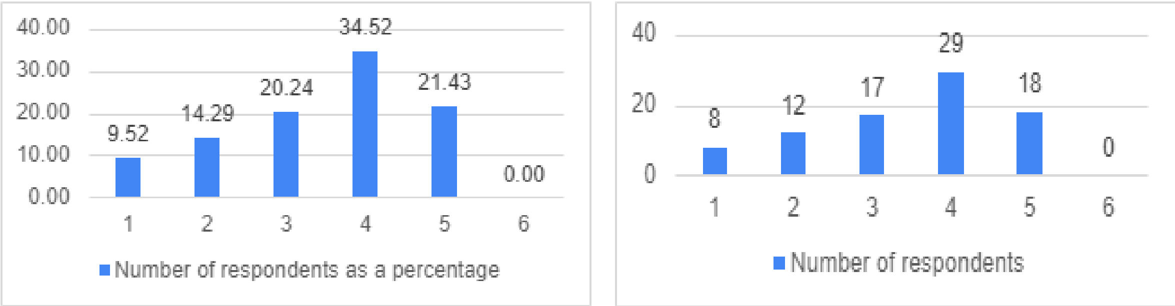


Figure 19 indicates the extent to which AI will be able to support the project manager in establishing and managing project teams. According to the above chart, the highest number of respondents mentioned that AI is capable of helping project managers develop and manage teams.

Figure 20 - Help of AI to assess the performance of team members.

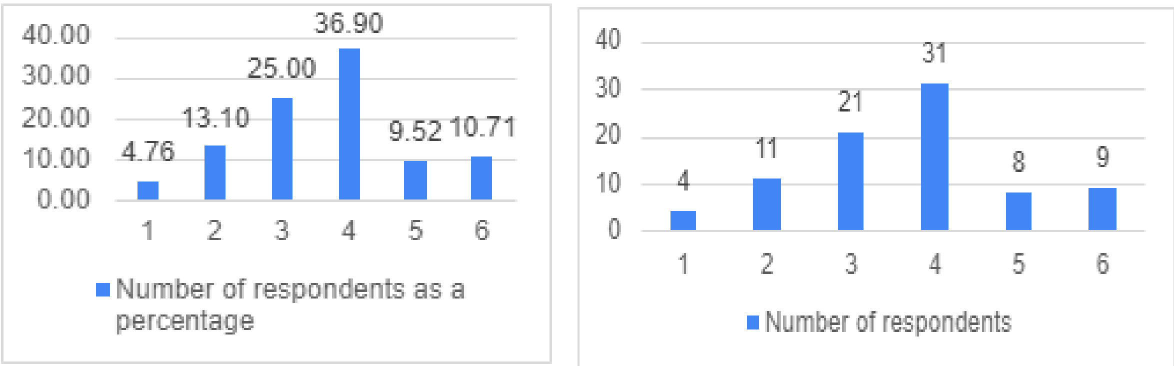


Figure 20 indicates the extent to which AI can help project managers assess the performance of team members. A higher number of respondents say that AI will help them to assess the performance of team members.

Referring to the above findings from Figures 15, 16, 17, 18, 19, and 21, **it disclosed that this analysis supported research objective 1**. (Objective 1: To find out whether there is an impact of AI integration on project management agility in Norwegian companies).

Figure 21 - What extent do the AI will affect project management tasks

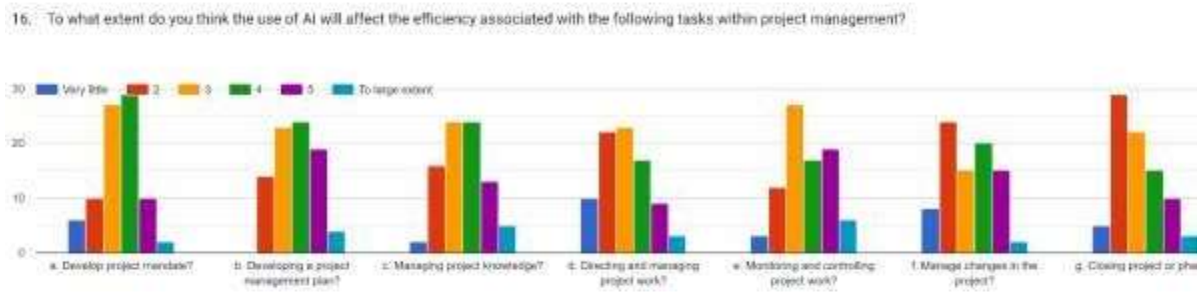


Figure 21 shows to what extent the project managers think the use of AI will affect the efficiency associated with the following tasks within project management.

Figure 22 - Type of training needed for the effective use of AI

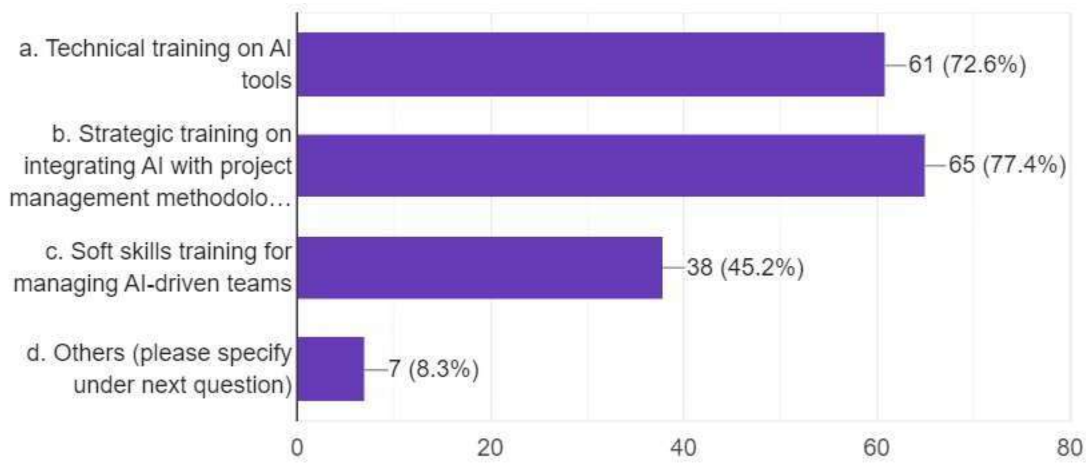


Figure 22 is about the types of training that the project managers would need to effectively use AI in project management. According to the above figure, the highest number of project managers said that there should be strategic training on integrating AI with project management methodologies. A large number of participants need technical training on AI tools, which is 72.6%, and 45.2% said there should be soft skills training for managing AI-driven teams.

In question 19 (Appendix – 1), respondents wrote about what other types of training will be needed and what challenges can occur during AI training for project managers. Here are some of the trainings and challenges mentioned,

1. The risk of data breaches and unauthorized access is a constant threat.
2. Understanding of the project managers regarding the advantages and disadvantages of AI use in project management.
3. One has mentioned the training time and highlighted that it would be better if they had time to familiarize themselves.
4. Talks about ethical considerations such as bias in algorithms.
5. Most people do not have enough knowledge to understand how to use AI efficiently in project management.

Investment in training and development is crucial for the successful deployment and use of AI in project management. Answers addressed possible challenges that can occur when utilizing AI for the industry of project management.

Figure 23 - The extent to which the usage of AI tools to manage projects

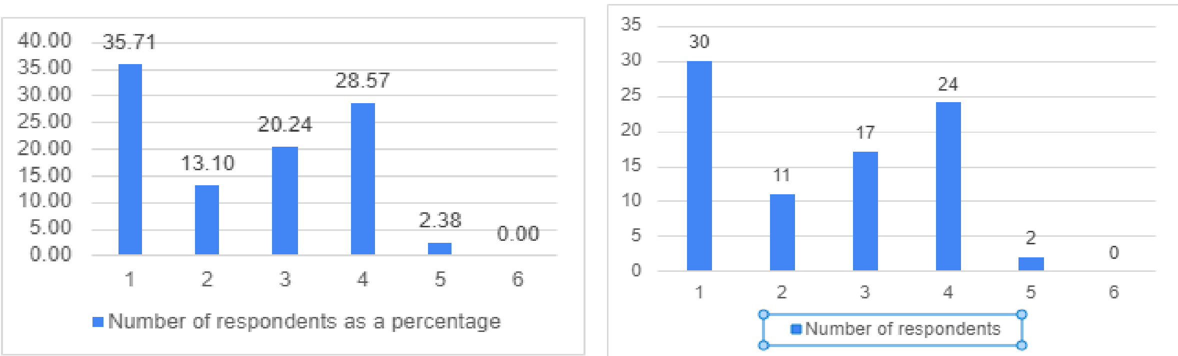


Figure 23 indicates the extent to which the respondents use AI-based tools to manage projects. A higher number of respondents mention that they do not use AI tools (35.71%).

Figure 24 - Usefulness of AI tools

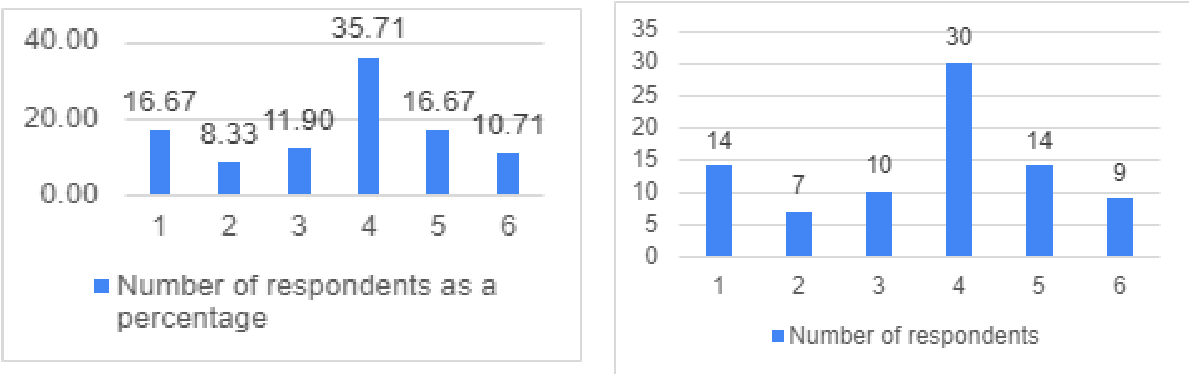


Figure 24 shows what the respondents think about the usefulness of AI tools. A higher percentage of respondents, which is about 35.71%, believe that AI tools are helpful. This indicates positive feedback to question 23 (Appendix – 1).

Figure 25 - Understanding the benefits of AI

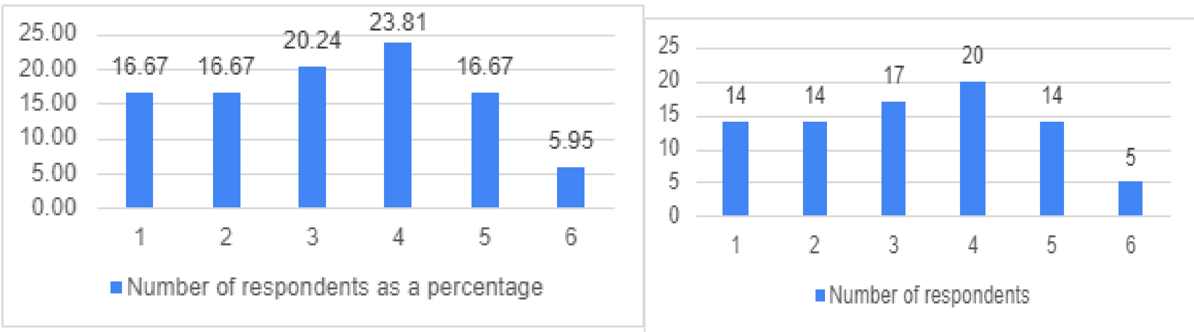
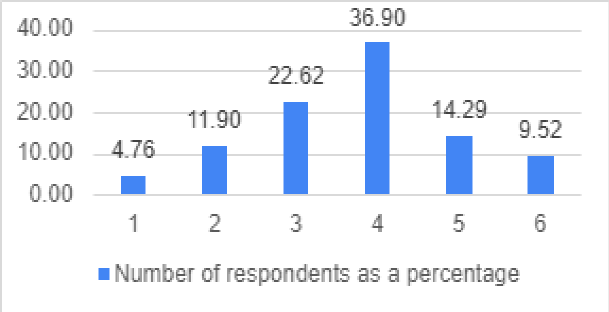


Figure 25 indicates that respondents have a nearly equal understanding of the benefits of AI. A higher number of respondents say they have some knowledge of the benefits.

Figure 26 - Adoption of new technology



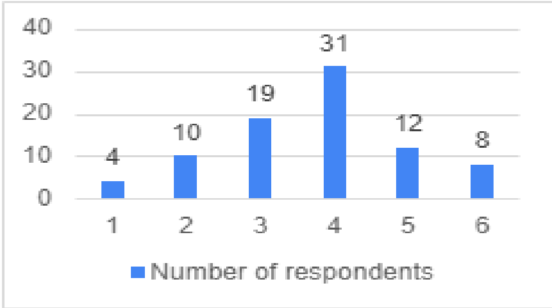
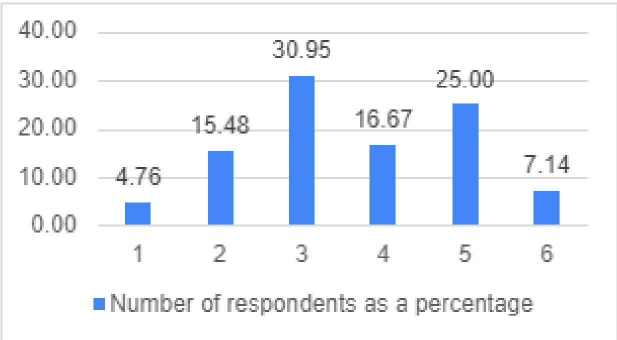


Figure 26 indicates the extent to which the readiness to adopt new technology for project management. The highest number of respondents indicated that their organizations are likely to adopt new technology, which is about 36.90%.

Figure 27 - The extent to which the company would like to wait to see the benefits of AI



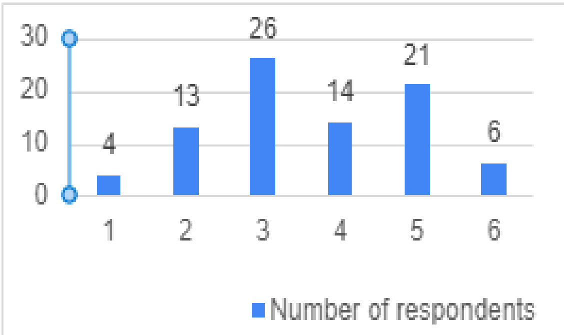


Figure 27 indicates the extent to which companies are willing to wait several months, up to a year, to start seeing the benefits of using AI. This is neutral.

Figure 28 - Usage of language models for project management

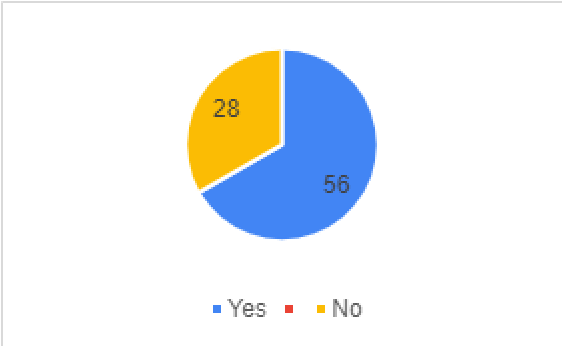


Figure 28 shows that 56% of respondents use language models for project management procedures.

Figure 29 - The extent to which the usage of language models

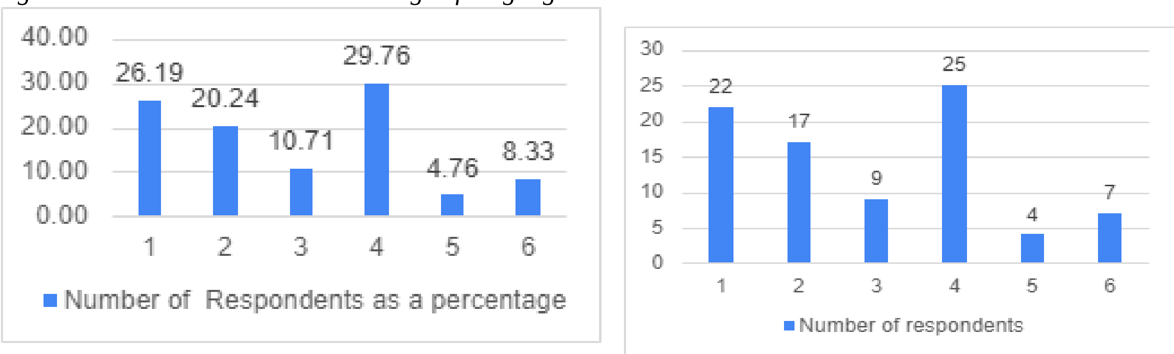


Figure 29 shows how the respondents use language models in project management. According to this, it shows that a lower number of respondents are using language models.

Figure 30 - Assessing the potential of using models

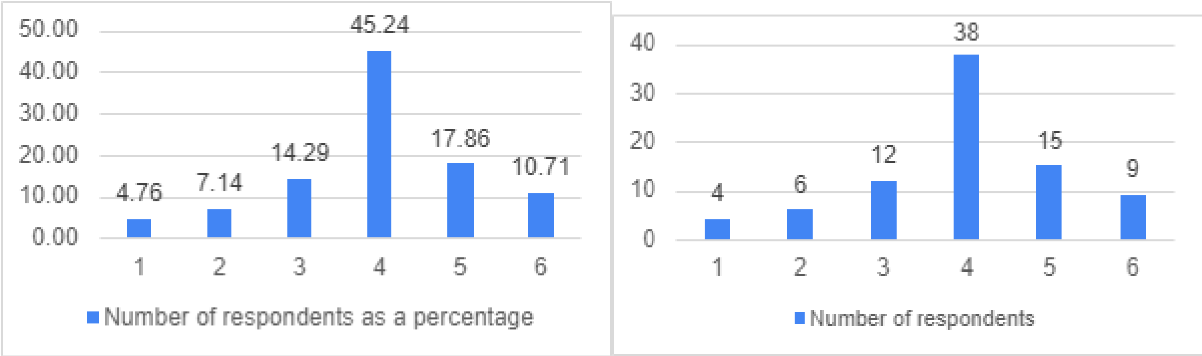


Figure 30 shows the extent to which the respondents assess the potential of using language models within project management. According to the graph, a higher number of respondents assess that there is a potential for using language models.

4.2.2 Reliability Analysis

Reliability analysis has been conducted to find the internal consistency and reliability of the measuring scales that are used to evaluate the variables under consideration, which include AI integration, Organizational learning culture, and Change management practices. The evaluation involved calculating Cronbach's Alpha, a statistic that shows how strongly a group of items are positively correlated. By using the appropriate questions for each variable, the researcher guaranteed the validity of the questions created to explain each group's variable. The analysis and interpretation of the findings are in Table 2 below:

Table 2 - Reliability Analysis

Variable	Cronbach's Alpha	No. of items
AI Integration	0.90	7
Organizational Learning Culture	0.74	3
Change Management Practices	0.82	3
Project Management Agility	0.76	4

Items represent the number of questions taken to analyze each variable.

The hypothetical Cronbach's Alpha values for organizational variables indicate generally strong reliability. Analysis indicates Cronbach's Alpha ranging from 0.74 to 0.90, which emphasizes acceptable levels for each variable. AI integration indicates Cronbach's Alpha value as 0.90, while Organizational learning culture indicates 0.74, exhibiting good consistency. Project management agility indicates 0.76, which demonstrates a higher level of consistency throughout the variables. Change management practices indicate good reliability, which is 0.82 among the variables that are used to assess consistency. Accordingly, all variables display acceptable reliability, suggesting some room for improvement in consistency.

These findings suggest that the measurement scales used to assess AI integration, Organizational learning culture, change management practices, and project management agility are reliable and consistent in measuring the intended constructs within the sample population. This indicates a higher value of Cronbach's Alpha and project management agility indicates that the variables in each scale are closely connected and efficiently measure the underlying constructs of interest.

4.2.3 Descriptive Analysis

Descriptive analysis provides an understanding of the characteristics within the study. The variables, namely AI integration, Organizational learning culture, change management practices, and Project management agility, were analyzed by using a sample of 84 project managers who were based in Norway.

Table 3 - Descriptive Analysis

Variable	N	Mean	Standard Deviation	Variance	Minimum	Maximum
AI Integration	**84**	3.43	0.93	0.86	2.00	5.14
Organizational Learning Culture	**84**	3.07	1.22	1.49	1.33	5.67
Change Management Practices	**84**	3.22	0.84	0.70	1.33	5.33
Project Management Agility	**84**	3.94	0.94	0.88	1.50	5.25

It demonstrates differences in mean values, indicating varying scores of integrations of AI for project management practices. The mean scores represent the average of each variable, AI integration, Organizational learning culture, Change management practices, and Project management agility, as 3.43, 3.07, 3.22, and 3.94. It demonstrates a medium to a higher level of AI adoption within the sample population, with an average AI integration value of 3.5. However, the range from 1.33 to 5.67 reflects substantial variability in AI adoption across different companies. This could be influenced by various independent variables, implying that AI adoption may enhance project management efficiency. In addition to this, values of standard deviation varied from 0.93 to 1.22, reflecting the level of dispersion variability of the scores around the mean.

Supportive organizational characteristics such as Organizational Learning Culture, Change Management Practices, and AI integration also show high mean values, suggesting that companies with solid internal practices are more likely to adopt AI effectively. Notably, the dependent variable, Project Management Agility, has a slightly higher mean, implying that AI adoption may enhance Project management efficiency.

4.2.4 Validity Analysis

Validity, in general, means the extent to which a test or measure is related to its intended purpose. For investigation choosing questionnaires, validity refers to the ability to sensitize the questionnaires for identification and detection of all symptoms under investigation (Rourke & Anderson, 2022). The KMO test is a test that has been designed with the aim of evaluating the appropriateness of data for factor analysis. In fact, it evaluates the appropriateness of the sample size. The test assesses the in-sample fit of each variable in the model and of the whole model (Shrestha, 2021).

The validity, for example, can be accessed via different methods. Table 6 represents the validity test's output generated through KMO and Bartlett's Test.

Table 6 – Validity Analysis

Variable	**KMO and Bartlett's Test**
AI Integration	0.86
Organizational Learning Culture	0.64
Change Management Practices	0.51
Project Management Agility	0.65

Considering Table 6, it is clear that all the KMO values generated after validity analysis are higher than the accepted validity level of 0.50. Therefore, it is clear that the instruments used in the current research are validated.

4.2.5 Normality Analysis

The normal data must be checked before any parametric tests are conducted. Normality assumption has to hold for the regression analysis to work accurately (Lumley et al., 2022). This representation indicates that the percentage simply undercuts, thus ensuring adequate analysis and the distribution of data. Each of the factors that the principal component analysis identified passes a routine normality check. The normality test was conducted by computing the skewness and kurtosis values. When the skewness and Kurtosis values are within the range of 0.2 to -0.2. It implies that the normality assumption is not violated (Demir, 2022).

Table 7 depicts the normal distribution of skewness and kurtosis scores in SPSS analysis.

Table 7 – Validity Analysis

Variable	Skewness Value	Kurtosis Value
AI Integration	0.154	-1.111
Organizational Learning Culture	0.478	-0.729
Change Management Practices	-0.024	-0.298
Project Management Agility	-0.350	-0.159

Considering Table 7, it is clear the fact that skewness and kurtosis values are within the acceptable range of 2 to -2. Therefore, the assumption of normality in the current research is not violated.

4.2.6 Correlation Analysis

Pearson's product-moment correlation coefficient is typically the parametric test analysis method. The Pearson correlation coefficient is applied to determine the rate of entanglement of two characteristics that are in metric format (Rahadian & Bandong, 2023). The correlation coefficient, or 'r' with a range from -1 to 1, tells the type of relation between two descriptive variables. Table 8 showcases the correlation output generated through SPSS analysis.

Table 7 –Correlation Analysis

Correlations				
Variable	PMA	AI-I	OLC	CMP
PMA	1	0.673**	0.755**	0.408**
AI-I		1	0.535**	0.219**
OLC			1	0.441**
CMP				1
Notes: **= Correlation is significant at the 0.01 level (2-tailed). PMA - Project Management Agility, AI-I – AI Integration, OLC – Organizational Learning Culture, Change Management Practices				

According to Table 7, there is a high degree of positive correlation between organizational learning culture with project management agility. There is a moderate degree of positive correlation between AI Integration and project management agility. Whereas there is a low degree of positive correlation between Change Management Practices the project management agility.

4.2.7 Regression Analysis

Generally, regression analysis is employed to assess how each of these independent variables affects the dependent variable of the Project Management Agility. Hence, it can be recognized which variables are vital for enhancing Project Management Agility and how these variables interact comprehensively with each other.

Moreover, regression analysis is a methodology that is used to test research hypotheses. In the current study, three research hypotheses (H1, H2, and H3) should be selected to test from the regression analysis. For example, if the regression analysis shows a positive number of the coefficient to AI integration, then it should be accepted as H1. (H1: There is a positive relationship between AI integration and project management agility.) Otherwise, it would be rejected.

In sum, this regression analysis shows whether AI integration and other variables are crucial, how important they are to one another, and how these variables contribute to the agility of project management in organizations in Norway. This analysis awareness is crucial for

Norwegian organizations for better enhancement of their project management procedures by focusing on the most significant factors.

Table 4 - Regression Analysis

Coefficients													
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	.622	.295		2.106	.038	.034	1.209					
	AI Integration	.383	.076	.379	5.062	.000	.233	.534	.673	.493	.320	.714	1.401
	Organizational Learning Culture	.389	.062	.508	6.231	.000	.265	.513	.755	.572	.394	.604	1.656
	Change Management Practices	.113	.079	.101	1.435	.155	-.044	.270	.408	.158	.091	.805	1.242
a. Dependent Variable: M_PMA													

It is clear that the p-value is less than the accepted beta value of 0.05. Therefore, align with the outcomes specified for Hypotheses 1 and 2 being accepted and Hypothesis 3 being rejected.

Considering the AI Integration, the P value (0.000) is less than 0.05, and the beta value is 0.379. It is denoted that when the AI integration increases in one unit, it is expected that the project management agility will increase by 0.38. Substantially impacts the agility of project management practices, confirming the critical role of AI in enhancing operational efficiency and responsiveness. This acceptance of Hypothesis 1 underscores H1: There is an impact of AI integration on project management agility in Norwegian companies.

Furthermore, Organizational Learning Culture also shows a significant positive impact (Beta value of 0.389 and p-value of 0.000), indicating that a learning-oriented organizational culture positively enhances the integration and effectiveness of AI in project management. This supports Hypothesis 2 (H2: There is an impact of organizational learning culture on project management agility in Norwegian companies), highlighting that there is an impact of organizational learning culture on project management agility in Norwegian companies.

On the other hand, Change Management Practices did not show a statistically significant effect (Beta value of 0.101 and p-value of 0.155), leading to the rejection of Hypothesis 3 (H3: There is an impact of change management practices on project management agility in Norwegian companies). This suggests that H3: There is an impact of change management practices on project management agility in Norwegian companies that do not significantly enhance the interaction between learning culture and AI integration in improving project management agility. Thereby, two hypotheses were accepted in the current research, and one was rejected.

Regression analysis provides insights into how AI integration is crucially supported by organizational learning, and it directly contributes to enhanced project management agility. However, change management practices in this scenario do not significantly influence this relationship, which may suggest a need for reevaluating the approach to change management in the context of AI deployment. Collectively, these results address Research Question 1 (RQ1: Does AI integration impact project management agility in Norwegian companies?) by confirming AI's significant impact on project management agility and providing indirect insights into Research Question 2 (RQ2: Does organizational learning culture impact project management agility in Norwegian companies?) suggesting that a supportive organizational culture likely fosters positive perceptions and interactions with AI among project team members and provide insights. Further, results provide insights with regard to the third question relating to change management practices is not aligned as it is insignificant.

4.3 Qualitative Analysis

4.3.1 Thematic Analysis

Thematic analysis has been done to examine the interview script (Caulfield, 2019), which investigates the integration and influence of AI technology in project management inside an organization. The study sought to uncover, analyze, and present patterns (themes) in the data. Six steps comprised the analytic process: becoming acquainted with the data, creating preliminary codes, looking for themes, evaluating themes, defining and recognizing themes, and producing a report.

Thematic Table

Table 5 - Thematic Table

Theme	Sub-themes	Illustrative Quotes
AI Integration in project Management	Roles and tasks of AI, Decision support	“AI has taken on specific roles such as…”
Impact of AI on efficiency and productivity	Contributions to projects, specific instances of impact	“AI integration has notably improved the efficiency by…”
Utilization of AI insights	Risk management, opportunities, areas for improvement	“Using AI, we’ve managed risks like…”
Satisfaction with AI integration	Current satisfaction levels suggested enhancements	“I am satisfied with our current AI integration because…”
Innovation and new ideas	Reception of innovations, implementation of new ideas	“New ideas are handled by…”
Change management in AI projects	Communication strategies, stakeholder involvement, handling resistance	“Change management is conducted by …”
Ethical Considerations	Fairness, security issues, transparency, job replacement and displacement	“Ethical issues can arrive from…”

Table 6 – Participants in the semi-structured interview

Participants	Gender	Education Level	Position
Interviewee 1	Male	Doctoral Degree (Ph.D)	Senior project manager
Interviewee 2	Male	Master’s Degree (MBA)	Project manager
Interviewee 3	Male	Master’s Degree	Project manager
Interviewee 4	Female	Master’s Degree	Project manager
Interviewee 5	Male	Doctoral Degree (Ph.D)	Senior project manager
Interviewee 6	Female	Master’s Degree	Project manager
Interviewee 7	Male	Master’s Degree	Project manager

The use of AI has made project management tasks easier, including scheduling, testing, and monitoring. I invited 20 project managers for the interview, and only seven people responded. Five of them are male, and 2 of them are female. This thematic analysis is based on the responses from the interviews that were conducted with project managers after they received responses to the questionnaire. This has come to identify the AI implementation in project management deeply.

AI integration in project management

Integration of AI within project management improves the effectiveness of project management activities (Wamba et al., 2020). Five of the interviews I conducted with project managers said (Interviewees 2, 3, 4, 6, 7) that they already use AI tools inside their company, and two (Interviewees 1 and 5) mentioned that they do not. The respondents who used those mentioned mainly use AI to automate their basic procedures, such as planning the project and allocating resources among team members, and those systems have the capability of monitoring real-time project updates. AI allows project managers to handle the team easily while reaching the desired outcome successfully. They mentioned that it gave them free time to work on other duties as well. Not only that, but it also increases precision in critical project decisions. As an example, AI technologies can forecast the risks that can happen in the future so project managers can take action to mitigate these risks.

Impact of AI on efficiency and productivity

AI is making a big contribution to the effectiveness of a project. Participants of the interview mentioned that AI has the capacity to identify the timeline of the project and allocate the resources fairly. These are based on the needed data at the beginning of the integration. Two interviewees mentioned that it would make them more accessible because of these technologies; they were able to finish their project within the exact time frame and the estimated cost. Another important thing is that AI can provide better communication facilities. So, team members can work together even if they are geographically not in one place.

Utilization of AI insights

AI utilization also talks about procedures for making decisions and managing the associated risks of the project. Some respondents mentioned that improvements need to be made in this area. They suggest that it would be better if there were some improvements to the risk prediction process.

Satisfaction with AI Integration

The satisfaction level of the AI use varied with respondents. Some mentioned that it has been very beneficial for them to perform their project management tasks, while some said that it is challenging because they are still not sure about the accuracy of the results. Two mentioned that they do not use AI for project management activities, but they would like to have it in the future if their organization allows them. Also, there are some suggestions from the respondents regarding the improvements in AI technologies and the accuracy of the outcome.

Innovation and new ideas

The receptiveness of the organization to new ideas and innovative approaches significantly influences the effectiveness of using AI in project management. The study highlights examples of how new project management methodologies, influenced by AI insights, have been adopted. The agility with which the organization incorporates new ideas correlates directly with its ability to stay competitive and effective in project delivery.

Ethical considerations

Respondents mentioned various ethical issues regarding the use of AI for project management. According to the respondents' feedback, bias is the main issue because the AI system is trained by feeding previous data and information, so it can be biased based on the data it already has when providing the output. It leads to unfair outcomes regarding the procedures of project management activities, such as dividing tasks among members and monitoring the individual performance of team members. So, respondents (Interviewees 1, 3,6) said that there is a risk regarding expecting a highly accurate outcome. AI needs a large amount of data, including very private information. So, privacy issues and security are other ethical issues mentioned by participants. Some respondents mentioned that the employees who work in their organizations have difficulties understanding AI tools due to the lack of transparency. Finally, the loss of occupations is a main issue when adapting AI to project management procedures.

Change Management in AI Projects

The management of change, particularly regarding AI project integration, requires careful planning and communication (Wamba-Taguimdje et al., 2020). Successful strategies include transparent communication about the benefits and impacts of AI, involving key stakeholders early in the process, and providing adequate support to address any resistance. Improvements could include more personalized communication strategies and greater involvement of team members in the planning stages to ensure buy-in and smoother transitions (Hengstler et al., 2016).

Overall, the thematic analysis reveals that while AI has significantly impacted project management processes, its full potential is yet to be realized. Continued investment in training, more robust integration of AI insights into strategic decision-making, and a proactive approach to change management are imperative. The organization can enhance its project management capabilities by addressing these areas in an AI-driven world.

5 Discussion, Recommendations, Limitations and Future Research

5.1 Discussion

This research focuses on how the utilization of the effect of AI on project management by indicating three independent variables (AI integration, Organizational learning culture, and change management practices) and One dependent variable (Project Management Agility). Technology advancement is an important element in management, and AI is capable of providing more insights into this field in the future with new technologies such as automation, machine learning, deep learning, and natural language processing (Ida et al. et al., 2021; Wijayati et al., 2022).

When considering the hypothesis assumed in this research, the study has discovered a positive correlation between AI integration and project management agility (H1: There is an impact of AI integration on project management agility in Norwegian companies). It has been proved in the regression analysis by indicating a coefficient of 0.383 and a p-value of 0.000. Results indicate that integrating AI into project management processes in the Norwegian context helps project managers to be more agile in managing their projects. The reason for supporting this relationship in the study could be related to the high level of technological advancement in Norway, which may have played a role in the openness of project managers to welcome new technological innovations. In addition, the Norwegian government is known to support initiatives related to technology. This includes funding for AI research and development, which serves as a strong foundation for AI integration. It is worth mentioning that several studies have drawn similar conclusions about this relationship. For example, using AI in project management positively affects the improvements of the project management procedures (Tominc et al., 2023a). Soravito (2020) and Mhlari (2023) mentioned that AI technologies regarding forecasting will help to identify dangers that can occur during the project timeline (Mhlari, 2020; Soravito, 2023). The finding implies a better response to changes in requirements, improved decision-making, and improved efficiency in project execution.

Next, the findings suggest that Organizational learning culture influences AI integration in project management agility (H2: There is an impact of organizational learning culture on project management agility in Norwegian companies) by showing a coefficient of 0.389 and p-value of 0.000 from the regression analysis (Table 6). Project managers can get advice from AI tools to increase their efficiency and can learn new processes (Taboada et al., 2023). AI's use of project management increases the learning and encouragement of organizations (Denning, 2016). Also, it was found that the change management approach does not have any effect on the interaction between Organizational learning culture and AI integration (H3: There is an impact of change management practices on project management agility in Norwegian companies) by indicating a coefficient of 0.113 and p-value of 0.155 from the regression analysis (Table 6). A study stated that utilizing AI in project management has a relationship with the change management process (Tominc et al., 2023b). There is no doubt that AI is capable of changing the way project management tasks and procedures are handled (Lahmann, 2018; Taboada et al., 2023).

When considering the research questions have been addressed in regression analysis by confirming that AI considerably impacts managing projects and organizational learning process and also positively impacts the project management agility within the project management context in Norway (Table 6). And 23.81% responded to number 4. This means the majority of respondents are ready to accept new technology and willing to wait several months to see the benefits of using AI (Figure 28). This means the project managers who live in Norway will be able to prepare before integrating AI into their companies. Respondents in the interview and for question 19 (Question 19 – Appendix) mentioned challenges such as some employees' resistance to change and shift from the old method of managing projects, the risk of data breaches, and unauthorized access. One mentioned that AI will sometimes give the wrong output.

When considering the research objectives, Objective 1 (Objective 1: find out whether there is an impact of AI integration on project management agility in Norwegian companies) has been addressed in frequency analysis by Figures 15, 16, 17, 18, 19, and 21. Also, Objective 2 (Objective 2: To find out whether there is an impact of organizational learning culture on project management agility in Norwegian companies) can be addressed by regression analysis by proving positively enhancing the integration of Organizational learning culture with AI in project management (Organizational learning culture shows Beta value of 0.389 and p-value of 0.003) (Table 4). Finally, change management practices did not show a statistically significant impact on project management agility in Norwegian companies (Change management practices Beta value of 0.113 and p-value of 0.155). Hence, change management practices do not have any impact on Project management agility in Norwegian companies (Objective 3: To find out whether there is an impact of change management practices on project management agility in Norwegian companies).

5.2 Recommendations

Firstly, Norwegian organizations can make investments to enhance project management processes. Shamim and Isham 2024 also mentioned that the integration of AI tools will increase the effectiveness of the project lifecycle, which enables smooth planning and resource allocation. Organizations can make their atmosphere facilitate the employees to learn, receive support, and share ideas with other employees while providing training regarding the establishment of new AI technologies (Edmondson, 2012). It increases the organization's adaptability according to changing requirements in a Norwegian context.

During the interview, some participants mentioned ethical considerations regarding the integration of AI. They were not sure about the accuracy of the results and the privacy of data needed to generate an outcome. So, organizations and practitioners need to be more careful and responsible about ethics when using AI for project management procedures (Lasaite, 2024). Finally, in the survey answers, almost every respondent mentioned that it is better to have training after the implementation of AI within Norwegian organizations, which will increase the effectiveness of using AI. This includes technical, strategic, and soft skills training. So, I can recommend that it is better to have an appropriate and adequate training program after eliminating new technologies in an organization.

5.3 Limitations and Future Studies

The small sample size of only 84 project managers in Norway is a primary limitation of this research. As a consequence, the validity of the outcome may restrict the generality. Also, the sample is originally from Norway; it does not represent a broader population. Because of this small sample size, it is a bit difficult to come to a highly effective conclusion. As Button et al. (2013) said, it may reduce the chance of identifying the actual effect of research. Because of this issue, future researchers can employ studies using a bigger sample and a more varied population, not only limiting it to one geographical region. Also, in this research, it took only three months to collect data, which is one reason for the small sample size. To avoid this, future researchers can use longitudinal studies, which collect data several times over a long period (Caruana et al., 2015). Then, future investigators will get a chance to examine the effects of the usage of AI in the field of project management in the long term and study the changes in AI adoption, project management agility, and organizational dynamics over time. Then can analyze the deeper relationship between variables. Case studies can also be employed to conduct an in-depth study regarding the subject area, and they provide significant understanding regarding standards, obstacles, and factors that contribute to success. Case studies allow academics to investigate the socioeconomic factors that influence AI adoption, organizational learning, management of change, and project results in real-world situations (Gerring, 2004).

Although attempts have been made to assure the accuracy of all measuring scales utilized in the research, limitations related to construct validity and measurement error may exist. Constructs such as AI integration, Organizational learning culture, and Project management agility are complex and multifaceted, making them challenging to measure accurately. Future research could refine measurement instruments and employ more robust validation techniques to enhance measurement validity. This study collected secondary data from published literature, which can lead to publication bias. Research that is rich with significant findings has a higher chance of getting published than research with ambiguous results. There is much research that has not been published in the English language or is unpublished. It may limit the comprehensiveness of the literature review. Future research could employ systematic review techniques to reduce publication bias and increase the rigor of the literature review process (Ganann et al., 2010).

The quick speed of technical innovation in AI and project management may render some findings outdated or less relevant over time—AI integration with the handling of project adaptability. Studies in the future should consider technological improvements and consider their implications for AI integration initiatives.

Future research needs to be more considerable about the ethical considerations, including concerns about privacy, bias, and responsibility of AI integration in project management. Future studies could explore stakeholders' perceptions, attitudes, and ethical considerations surrounding AI adoption in project management and develop frameworks for responsible AI use (Lasaite, 2024). It is important to do further research on advanced AI applications. This includes predictive analysis, automated project management systems, and cognitive computing. Because this research does not discuss these technologies in depth, it mentions AI as a whole, not specifically technologies inside AI. Future research can study advanced AI applications to receive more reliable and accurate outcomes.

Nagireddy (2023) addressed that the project management community has been relatively quiet about embracing AI approaches, even though these techniques may give several benefits to project managers. Future studies should dive deeper into the variables that drive most organizations to be cautious about adopting new technologies. To solve questions such as, "How does the project management team know the required data?" and "When should they use it?" Further research on the relevant subject is required. Based on these findings, more academic research into AI applications, such as project management, is required (Nagireddy, 2023). Finally, collaborative research can be employed in the future to gain a reliable outcome. It encompasses such stakeholders as academics, representatives of the industry, and policymakers who might foster multidisciplinary research, information sharing, and technology transfer in the field of AI integration and project management. Collaboration could be used to speed up the transition of research findings into practice and make it possible to look into real-world problems emerging in organizing activities (Katz & Martin, 1997).

6 Conclusion

This gave valuable insight into how AI Integration affects a project management team and considers the organization's overall performance. On the relation between the integration of AI, Organizational learning culture, Change management practices and Project management agility, some remarkable findings come out. To begin with, the research outcomes prove that AI application contributes to a higher level of project management agility. This demonstrates that an organization is able to adapt to any new requirement previously. This positive relationship was investigated by Mahmood et al. (2023). Apart from these, organizational learning culture is an important factor regarding the integration of AI technologies, which supports the integration and adoption inside an organization (Jarrahi et al., 2022).

Furthermore, it found that change management practices do not have any influence on the improvement of the connection between AI integration and Organizational learning culture. Additionally, results indicated the connection between AI integration and agility of project management. Of the respondents who answered the questionnaire and were interviewed, a higher number of them are using AI technologies. Participants from the IT services and products mentioned that they are already using some AI technologies to create project timelines and schedule and monitor team members, which was proved by indicating a higher percentage for Question number 3 (Appendix 1) in the survey. Two participants from the interview mentioned that they like to use AI tools for their project management procedures, but their organizations do not allow them to use those. In the interview, a few (2 interviewees) mentioned that they were not sure about these technologies and believed that they could get a better output by conventional methods. Respondents suggest that it would be good if there were an improvement in the accuracy of the result and ethical considerations.

Overall, the identified result of this study emphasizes the possibilities of using AI to manage projects and organizational agility (Fosso Wamba, 2022). It enables organizations to improve their project management agility while paving the way to achieve the desired goal within various business environments (Patrício et al., 2021).

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8 Appendices

8.1 Appendix 1 – Questionnaire

Kan KI brukes til å gjøre prosjektledelse mer smidig?

Form description

1. Hvor gammel er du?

- 18 – 30
- 31 – 40
- 41 - 50
- 51 - 60
- Over 60

2. Hva er ditt kjønn? *

- Mann
- Kvinne
- Ønsker ikke å svare

3. I hva slags virksomhet jobber du? *

- Finansielle tjenester (f.eks. bank, forsikring)
- IT-produkter og tjenester
- Helsevesen og legemidler
- Forbruksvarer og detaljhandel (f.eks. elektronikk, klær)
- Produksjon og industri (f.eks. fødevarer, maskiner)
- Telekommunikasjon
- Transport
- Offentlig sektor
- Ikke-kommersiell organisasjon
- Annen

4. Hva er din rolle i selskapet? *

- ◯ Leder på høyere nivå (CEO, CFO, COO, CIO, etc.)
- ◯ Leder på mellomnivå
- ◯ Har ikke lederansvar

5. Hvor mange år har du jobbet med prosjektledelse? *

- ◯ 0 - 2
- ◯ 3 - 5
- ◯ 5 - 10
- ◯ Mer enn 10

6. Bruker organisasjonen din KI-baserte systemer? *

- ◯ Ja
- ◯ Ikke ennå (fremtidsplan)
- ◯ Jeg vet ikke
- ◯ Nei

7. Hvordan er kunnskapen din om KI i prosjektledelse? *

- Veldig bra
- Ganske bra
- Bevisst om muligheter
- Begrenset forståelse
- Ikke-eksisterende

8. Mener du at KI-baserte systemer vil kunne hjelpe prosjektledere til økt effektivitet ifm. administrasjon av prosjekter (nå milepæler/tidsfrister)? *

- Ja
- Nei
- Jeg vet ikke

9. Hva synes du om KI i prosjektledelse? *

- ◯ Redd
- ◯ Bekymret
- ◯ Fasinert
- ◯ Spent
- ◯ Ikke interessert

10. Vil bruk av KI påvirke prosjektlederrollen? *

- ◯ Ikke i det hele tatt
- ◯ Sannsynligvis
- ◯ Muligens
- ◯ Definitivt
- ◯ Jeg vet ikke

11. Vennligst del dine personlige tanker om KI i prosjektledelse? (Åpent spørsmål)

Long answer text

12. I hvilken grad tror du KI vil bidra til mer effektiv prosjektledelse? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

13. Tror du KI vil kunne hjelpe til å sette sammen team og foreta delegering av oppgaver og plikter til hvert teammedlem? *

○ Ja

○ Nei

○ Jeg vet ikke

14. Hvis ja, i hvilken grad tror du KI vil kunne støtte prosjektlederen i arbeidet med å etablere og administrere prosjektteam? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

15. I hvilken grad du tror KI kan hjelpe projektledere med å vurdere resultatene til teammedlemmer ? (hjelp med å identifisere områder hvor ytterligere opplæring eller støtte kan være nødvendig) *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

16. I hvilken grad tror du bruk av KI vil påvirke effektiviteten knyttet til følgende arbeidsoppgaver innen prosjektledelse? *

	1 Veldig liten	2	3	4	5	6 I stor grad
Utvikle prosj...	○	○	○	○	○	○
Utvikle en pr...	○	○	○	○	○	○
Administrere...	○	○	○	○	○	○
Styre og led...	○	○	○	○	○	○
Overvåke og...	○	○	○	○	○	○
Håndtere en...	○	○	○	○	○	○
Avslutte pro...	○	○	○	○	○	○

17. Hvordan ser du for deg at KI påvirker arbeidsflyter for prosjektledelse ? (Åpen)

Short answer text

18. Hvilken type opplæring tror du prosjektledere trenger for å effektivt bruke KI i prosjektledelse? (Velg alt som gjelder) *

- [ ] Teknisk opplæring i KI-verktøy
- [ ] Strategisk trening på integrering av KI med prosjektledelsesmetodikk
- [ ] Myk ferdighetstrening for å administrere KI-drevne team
- [ ] Andre (vennligst spesifiser under neste spørsmål)

19. Hvilke utfordringer ser du for deg ved å tilby KI-opplæring for prosjektledere? (Åpen)

Long answer text

20. Bruker du eller andre ansatte i virksomhetene KI-baserte verktøy for å administrere prosjekter? *

- ◯ Ja
- ◯ Nei
- ◯ Jeg vet ikke

21. Hvis ja, i hvilken grad bruker du eller andre ansatte KI-baserte verktøy for å administrere prosjekter? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

22. I hvilken grad bruker du personlig slike verktøy? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

23. I hvilken grad synes du slike verktøy er nyttige? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

24. Hvor trygg er du på din forståelse av de potensielle fordelene med KI for prosjektledelse? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

25. I hvilken grad er din organisasjon klar til å akseptere og ta i bruk ny teknologi for prosjektadministrasjon? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

26. I hvilken grad er virksomhetens ledelse villig til å vente i flere måneder, opptil ett år, for å begynne å se fordelene med bruk av KI? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

27. Bruker du allerede språkmodeller (ChatGPT eller lignende) i prosjektledelse? *

- ○ Ja
- ○ Nei

28. I hvilken grad bruker du noen språkmodell (ChatGPT eller lignende) innen prosjektledelse? *

	1	2	3	4	5	6	
veldig liten	○	○	○	○	○	○	I stor grad

29. Hvordan vurderer du potensialet ved bruk av språkmodeller (ChatGPT eller lignende) innen prosjektledelse? *

	1	2	3	4	5	6	
Veldig liten	○	○	○	○	○	○	I stor grad

8.2 Appendix 2 – Interview Questions

1. Can you describe how AI technology is currently integrated into your project management processes? What specific roles or tasks has AI taken on?
2. In what ways has AI technology directly assisted in decision-making within your project management activities? Could you provide specific examples or outcomes from this integration?
3. Reflecting on recent projects, how would you describe the contribution of AI integration to the overall efficiency and productivity of these projects? Can you cite specific instances where AI made a discernible difference?
4. How effectively do you think the organization utilizes AI-driven insights to anticipate and manage project risks and opportunities? Are there areas where this could be improved?
5. Overall, how satisfied are you with the level of AI integration in your project management practices, and why? What changes would you suggest enhancing this integration?
6. How does the organization encourage and facilitate continuous learning and knowledge sharing among project teams? Can you discuss any specific programs or initiatives that have been particularly impactful?
7. How receptive is the organization to implementing new ideas and innovative approaches in project management? Can you give examples of how new ideas were handled in recent projects?
8. In your experience, how effectively does the organization capture and disseminate lessons learned from past projects? Are there mechanisms you think could improve this process?
9. Looking at change management, particularly with AI integration projects, how well does the organization communicate changes, involve stakeholders, and manage any resistance? What strategies have worked well, and what could be improved?

10. From your perspective, what ethical considerations arise from the use of AI in project management?

8.3 Appendix 3 – SPSS Output

Descriptive Statistics

Statistics

		M_AI_I	M_OLC	M_CMP	M_PMA
N	Valid	84	84	84	84
	Missing	0	0	0	0
Mean		3.4252	3.0714	3.2222	3.4940
Median		3.4286	3.0000	3.3333	3.7500
Mode		2.29	1.67	3.33	3.75
Std. Deviation		.92581	1.22236	.83647	.93620
Variance		.857	1.494	.700	.876
Minimum		2.00	1.33	1.33	1.50
Maximum		5.14	5.67	5.33	5.25

Reliability Anal3ysis

AI Integration

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.907	.908	7

Item Statistics

	Mean	Std. Deviation	N
AI Integration 1	3.46	1.011	84
AI Integration 2	3.71	1.136	84
AI Integration 3	3.56	1.165	84
AI Integration 4	3.14	1.153	84
AI Integration 5	3.69	1.212	84

AI Integration 6	3.29	1.218	84
AI Integration 7	3.12	1.176	84

Organizational Learning Culture

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.738	.742	3

Item Statistics

	Mean	Std. Deviation	N
Organizational Learning Culture 1	2.92	1.585	84
Organizational Learning Culture 2	3.65	1.275	84
Organizational Learning Culture 3	2.64	1.640	84

Change Management Practices

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.817	.806	3

Item Statistics

	Mean	Std. Deviation	N
Change Management Practices 1	2.86	1.346	84
Change Management Practices 2	3.18	1.174	84
Change Management Practices 3	3.63	1.333	84

Project Management Agility

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.759	.762	4

Item Statistics

	Mean	Std. Deviation	N
Project Management Agility 1	4.08	1.143	84
Project Management Agility 2	3.44	1.245	84
Project Management Agility 3	2.49	1.303	84
Project Management Agility 4	3.96	1.217	84

Validity Analysis

AI Integration

KMO and Bartlett's Test[a]

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.858
Bartlett's Test of Sphericity	Approx. Chi-Square	370.214
	df	21
	Sig.	.000

a. Based on correlations

Organizational Learning Culture

KMO and Bartlett's Test[a]

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.643
Bartlett's Test of Sphericity	Approx. Chi-Square	59.172
	df	3
	Sig.	.000

a. Based on correlations

Change Management Practices

KMO and Bartlett's Test[a]

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.501
Bartlett's Test of Sphericity	Approx. Chi-Square	9.103
	df	3
	Sig.	.028

a. Based on correlations

Project Management Agility

KMO and Bartlett's Test[a]

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.648
Bartlett's Test of Sphericity	Approx. Chi-Square	93.987
	df	6
	Sig.	.000

a. Based on correlations

Testing Normality

Descriptive Statistics

	N	Skewness		Kurtosis	
	Statistic	Statistic	Std. Error	Statistic	Std. Error
M_AI_I	84	.154	.263	-1.111	.520
M_OLC	84	.478	.263	-.729	.520
M_CMP	84	-.024	.263	-.298	.520
M_PMA	84	-.350	.263	-.519	.520
Valid N (listwise)	84				

Correlation Analysis

Correlations

		M_PMA	M_AI_I	M_OLC	M_CMP
M_PMA	Pearson Correlation	1	.673**	.755**	.408**
	Sig. (2-tailed)		.000	.000	.000

	N	84	84	84	84
M_AI_I	Pearson Correlation	.673**	1	.535**	.219*
	Sig. (2-tailed)	.000		.000	.046
	N	84	84	84	84
M_OLC	Pearson Correlation	.755**	.535**	1	.441**
	Sig. (2-tailed)	.000	.000		.000
	N	84	84	84	84
M_CMP	Pearson Correlation	.408**	.219*	.441**	1
	Sig. (2-tailed)	.000	.046	.000	
	N	84	84	84	84

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Hypothesis Testing

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.824[a]	.680	.668	.53984	.680	56.542	3	80	.000	2.077

a. Predictors: (Constant), M_CMP, M_AI_I, M_OLC

b. Dependent Variable: M_PMA

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	49.433	3	16.478	56.542	.000[b]
	Residual	23.314	80	.291		
	Total	72.747	83			

a. Dependent Variable: M_PMA

b. Predictors: (Constant), M_CMP, M_AI_I, M_OLC

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	95.0% Confidence Interval for B Lower Bound	95.0% Confidence Interval for B Upper Bound	Correlations Zero-order	Correlations Partial	Correlations Part	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	.622	.295		2.106	.038	.034	1.209					
	M_AI_I	.383	.076	.379	5.062	.000	.233	.534	.673	.493	.320	.714	1.401
	M_OLC	.389	.062	.508	6.231	.000	.265	.513	.755	.572	.394	.604	1.656
	M_CMP	.113	.079	.101	1.435	.155	-.044	.270	.408	.158	.091	.805	1.242

a. Dependent Variable: M_PMA

8.4 Appendix 4: List of Abbreviations

PM - Project Management

AI - Artificial Intelligence

ML - Machine Learning

DP - Deep Learning

NLP - Natural Language Processing

ANN - Artificial Nural Network

NN - Nural Networks

ASI - Artificial Super Intelligence

AGI - Artificial Generic Intelligence

IDC - International Data Corporation

OECD - Organization for Economic Co-operation and Development

ICT - Information and Communication Technology

R&D - Research and Development

PMBOK - Project Management Body of Knowledge

SPM - Software Project Management

PMB - Project Management Bots

MADM - Multiple Attribute Decision Making

EPSM - Evolutionary Project Success Prediction Model

SVM - Support Vector Machine

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